Catena 187 (2020) 104353

Contents lists available at ScienceDirect

Catena

journal homepage: www.elsevier.com/locate/catena

Effects of soil and vegetation development on surface hydrological properties of moraines in the Swiss Alps

Fabian Maier[a,*], Ilja van Meerveld[a], Konrad Greinwald[b], Tobias Gebauer[b], Florian Lustenberger[a], Anne Hartmann[c], Alessandra Musso[a]

[a] Department of Geography, University of Zurich, Winterthurerstrasse 190, 8057 Zurich, Switzerland
[b] Department of Geobotany, Faculty of Biology, Albert-Ludwigs University Freiburg, Schänzlestrasse 1, D-79104 Freiburg, Germany
[c] GFZ German Research Centre for Geosciences, Telegrafenberg, D-14473 Potsdam, Germany

ARTICLE INFO

Keywords:
Saturated hydraulic conductivity
Hillslope age
Co-evolution
Preferential flow
Soil texture

ABSTRACT

The near-surface saturated hydraulic conductivity (K_{sat}) is an important hydrological characteristic because it determines surface infiltration rates and the vertical and lateral redistribution of water in the soil. However, there is comparatively little knowledge about the changes in K_{sat} during landscape development and how the co-evolution of biological, pedological and hydrological characteristics affect the movement of water through the soil. On the one hand, increasing vegetation cover is expected to increase macroporosity and thus K_{sat}. On the other hand, clay formation is expected to decrease K_{sat}. To investigate how hillslope aging affects K_{sat}, we used a space-for-time approach and conducted comprehensive measurements of vegetation, soil and topography on a chronosequence of moraines in a proglacial area of the Swiss Alps. On four moraines, ranging from about ten thousand to ~30 years in age, we measured for three plots the near-surface soil characteristics. Surface and near-surface K_{sat} were high and decreased with depth on all moraines. Surface K_{sat} was highest on the youngest moraine (median: 4320 mm hr^{-1}) and lowest (540 mm hr^{-1}) on the oldest moraine. K_{sat} was significantly positive correlated with soil texture and the gravel content in the surface soil layer. The correlation analyses and Structural Equation Model suggested that the larger fraction of small particles for the older moraines had a bigger effect on K_{sat} than the denser root network. Even though the variability in measured K_{sat}-values within the moraines was high and water movement is thus likely very heterogeneous, the measured K_{sat} values suggest that infiltration-excess overland flow is very unlikely on these hillslopes but (lateral) near surface flow likely increases with the age of the hillslope. This information is important for understanding differences in runoff generation mechanisms in alpine areas with moraines of different ages, as well as landscape evolution models.

1. Introduction

Near surface hydrological conditions determine infiltration rates (Mayor et al., 2009; Ruggenthaler et al., 2016; Scherrer et al., 2007), overland flow generation and surface hydrologic connectivity (Cammeraat and Imeson, 1999; Cerdà, 1997; Huang et al., 2001) and soil erosion (Bryan, 2000; Fitzjohn et al., 1998). However, the near surface conditions are also affected by these processes. These feedbacks are referred to as co-evolution (Troch et al., 2013) and shape the characteristics of our landscapes and watersheds.

The saturated hydraulic conductivity (K_{sat}) is one of the most important surface hydrological characteristics because it determines the partitioning of water at the surface into overland flow and infiltration. It thus affects streamflow generation and erosion during high intensity rainfall events (Archer et al., 2013; Zimmermann et al., 2013). Similarly, the near surface K_{sat} determines the lateral and vertical distribution of the water within the soil, which in turn affects the distribution of soil moisture and thus plant available water (Rudner, 2005; Tromp-van Meerveld and McDonnell, 2006), soil development and weathering rates (Egli et al., 2015; Kelly et al., 1998). However K_{sat} is also influenced by the vegetation, soil development and weathering. Numerous studies have investigated how K_{sat} is affected by soil properties, such as clay content and bulk density (Elhakeem et al., 2018; Papanicolaou et al., 2015), vegetation and microtopography (Bedford and Small, 2008; Dunkerley, 2000; Lyford and Qashu, 1969; Nicolau et al., 1996), land use (Bormann and Klaassen, 2008; Zhang et al., 2018; Zimmermann et al., 2006; Zwartendijk et al., 2017), or climate (Elhakeem et al., 2018; Robinson et al., 2019). K_{sat} is generally smaller

* Corresponding author.
E-mail address: fabian.maier@geo.uzh.ch (F. Maier).

https://doi.org/10.1016/j.catena.2019.104353
Received 20 August 2019; Received in revised form 30 October 2019; Accepted 6 November 2019
Available online 05 December 2019
0341-8162/

in poorly sorted and fine textured soils than in coarser soils. This relation has led to the development of pedotransfer functions that predict K_{sat} based on soil texture, porosity and bulk density (Mbonimpa et al., 2002; Schaap et al., 2001; Tietje and Hennings, 1996; Wösten et al., 2001). However, in fine textured soils, soil structure is more important than soil texture and the majority of the water flows through preferential flow pathways (Beven, 2018; Weiler, 2017), so that K_{sat} is mainly affected by vegetation and soil fauna. For example, several studies have shown that K_{sat} can decrease rapidly with forest clearing due to compaction and a decrease in preferential flow pathways (Alegre and Cassel, 1996; Germer et al., 2010; Martinez and Zinck, 2004; Ziegler et al., 2004) but that surface K_{sat} can also recover within a few years after reforestation due to the increase in soil organic carbon content and root density (Ghimire et al., 2013; Zhang et al., 2019; Zimmermann et al., 2010; Zwartendijk et al., 2017). Seasonal changes in K_{sat} have been observed in maize fields due to plant growth and the emergence of root-driven soil macropores and preferential flow pathways (Prieksat et al., 1994; Wang et al., 2016). Climate change is expected to increase the macroporosity in some regions and therefore to affect streamflow responses (Hirmas et al., 2018).

However, so far little is known about the changes in K_{sat} or other near surface characteristics during landscape evolution, even though this affects modeled erosion rates and thus topography, as well as water redistribution in the soil, vegetation patterns and weathering, and thus landscape evolution. Several studies on mining piles (Evans et al., 2015; Pellegrini et al., 2016) and artificially created catchments, particularly the Chicken Creek Catchment (Gerwin et al., 2009; Schaaf et al., 2013), have shown that surface hydrological characteristics can change drastically during initial landscape development. For example, initial crusting of unvegetated mine soils led to lower infiltration rates and subsequently higher overland flow and erosion than expected based on the soil texture (Holländer et al., 2009), but this decreased rapidly once vegetative cover was established (Clark and Zipper, 2016; Moreno-de las Heras et al., 2009). Schaaf et al. (2017) found for the Chicken Creek Catchment that initially an abiotic-abiotic feedback mechanism (between substrate properties and water) dominated early landscape evolution, but that later abiotic-biotic processes (between substrate properties, water and vegetation) became more important and increased water storage, soil moisture and lateral subsurface flow in the unsaturated zone.

In order to study longer term changes in hillslope or catchment characteristics, the scientific community has to use a chronosequence or 'space-for-time substitution' approach (Bernasconi et al., 2011; Burga et al., 2010; Evans, 1999). Chronosequences have been used to study the changes in soil properties (D'Amico et al., 2015; Egli et al., 2010; Evans, 1999; Johnson et al., 2015; Vilmundardóttir et al., 2014) and plant succession (Burga et al., 2010; D'Amico et al., 2014, 2015; Helm et al., 1995; Mori et al., 2008). They are also used to investigate the role of climate and anthropogenic forcing on the hillslope properties (Eppes et al., 2008) or to analyze the co-evolution of bedrock weathering and topography (Rasmussen et al., 2017). However, despite these efforts to improve the understanding of long-term changes in hillslope surface properties, so far there is very limited information from chronosequences on the changes in near surface hydrological characteristics.

On the one hand, K_{sat} is expected to increase during landscape evolution due to increased vegetation cover and associated increases in soil organic matter content and the development of macropores and preferential water flow pathways (Beven and Germann, 2013; Weiler and Flühler, 2004; Wu et al., 2017; Zhao et al., 2013). On the other hand, the vegetation may also lead to more water-repellent topsoils that hamper surface infiltration (Mao et al., 2016). Furthermore, weathering and clay development and eluviation, may cause a lower K_{sat} for the subsoil. Lohse and Dietrich (2005), for example, found that K_{sat} was lower for older soils developed on lava flows than younger soils because of the higher clay content. For the older soils these lower permeability layers were assumed to promote ponding of water and increase lateral subsurface flow, as well as potentially saturated overland flow. In contrast, younger more porous soils that had not undergone significant soil formation yet, did not have these lower permeability layers and had a higher K_{sat}, promoting deeper percolation. Jefferson et al. (2010) suggested that the higher peak flows and lower baseflows for the catchments in the older Low Cascades in Oregon compared to catchments in the younger High Cascades were caused by lateral flow due to the higher clay content of the older soils in the Low Cascades. Young et al. (2004) measured K_{sat} at different depths on a chronosequence of desert soils, ranging from 50 to 100,000 years in age and similarly showed a decrease in K_{sat} of the surface layer due to the development of A-horizons with a clay-rich soil texture. However, other studies did not identify soil texture as a dominant driver for K_{sat}. Jarvis et al. (2013) reviewed 753 individual data-sets for 144 different locations worldwide and did not find a clear relation between soil texture and top-soil K_{sat}. Instead they found that K_{sat} was mainly related to bulk density, organic carbon content and land use. Similarly, Fischer et al. (2015) found that K_{sat} was not related to soil texture but was instead related to plant species richness. Liu et al. (2019) found for a semi-arid grassland that K_{sat} increased with species diversity. Thompson et al. (2010) analyzed field data from hardwood, pine and grassland sites in North Carolina combined with a dataset of nearly 50 vegetation communities across climatic gradients ranging from hyperarid deserts to the humid tropics and found similar significant positive correlations between aboveground biomass and K_{sat} in water-limited ecosystems.

Despite the large number of studies on K_{sat} at individual sites, investigations of the temporal changes in K_{sat}, particularly over long time scales (> 1000 years) are rare and usually do not consider the feedback processes that are involved in landscape development, particularly the feedback between soil and vegetation development. Furthermore, most of the studies on the effects of landuse on K_{sat} focused on tropical soils or desert soils. There is almost no data on K_{sat} of moraines in high alpine areas. We, therefore, conducted measurements of the near surface hydrological characteristics along a chronosequence of moraines in a glacier forefield of the Swiss Alps to answer the following questions:

- How do surface and near surface K_{sat} change during the first 10,000 years of soil and vegetation development on moraines?
- Is K_{sat} affected more by vegetation development or changes in soil texture?

We hypothesized that the increase in vegetation and associated increases in root length, root density and clay content would result in more stable soil aggregates (Hudek et al., 2017a; Lado et al., 2004) and macropores (Bundt et al., 2001; Johnson and Lehmann, 2016), which would have a larger effect on K_{sat} than weathering and clay accumulation. While the main focus of the study was on the surface hydrological characteristics of the moraines, we also report the K_{sat} values for the near surface layers because they affect the partitioning of the infiltrating water into lateral subsurface flow and deeper vertical percolation.

2. Study site

The field measurements were conducted in the forefield of the Steingletscher, south of the Sustenpass mountain road (~47°43′N, 8°25′E) in the Central Swiss Alps (Fig. 1a). The elevation of the forefield ranges between 1800 and 2100 m above sea level (asl). The overall orientation of the forefield is north-northwest. The area is characterized by an Alpine climate. At the Grimsel Hospiz, the closest long term automatic weather station (1979 m asl; located 20 km from the study area), the mean annual air temperature is 1.9 °C (1981–2010 period; MeteoSwiss, 2016a); the average monthly temperature varies between −5.4 °C in February and 10.0 °C in July. The precipitation is evenly distributed over the year; mean annual precipitation is 1856 mm y^{-1} (1981–2010 period; MeteoSwiss, 2016a). Snowfall events are common,

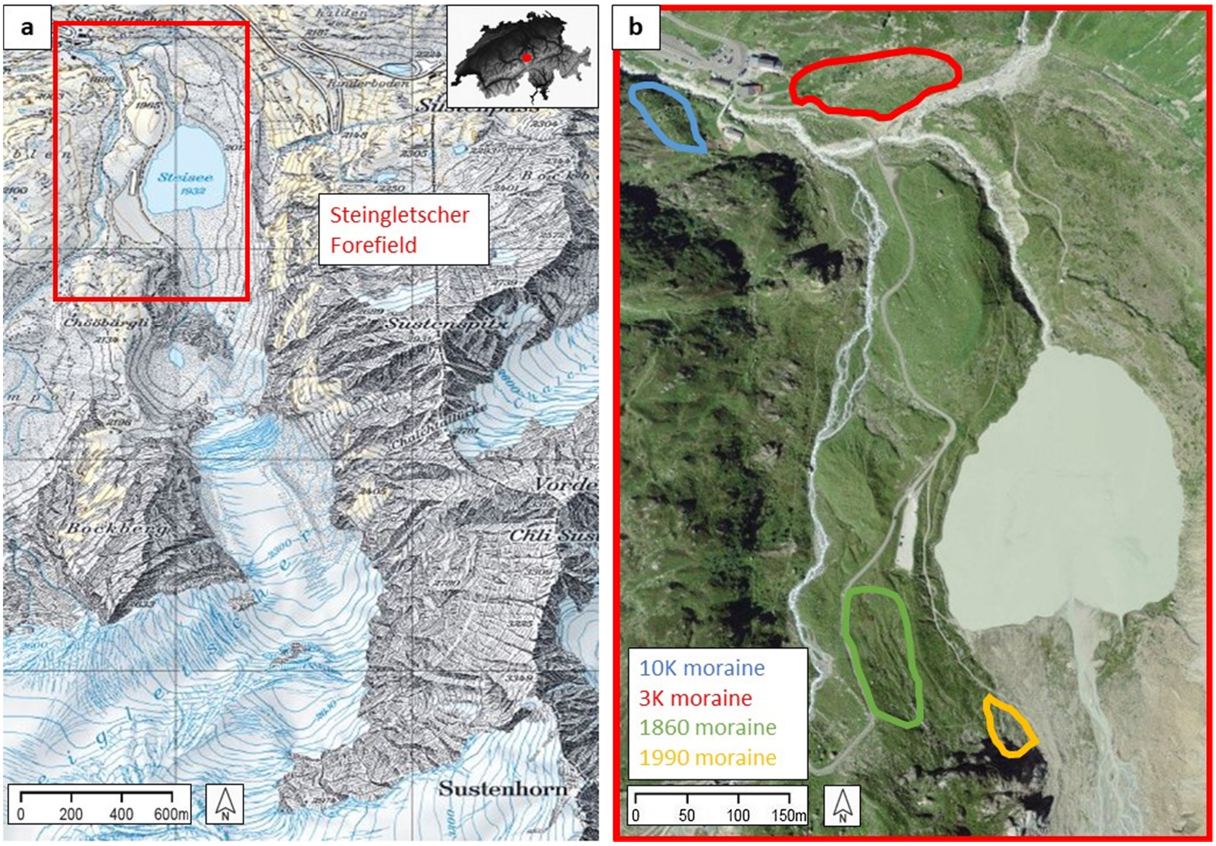


Fig. 1. Map (left) and aerial image (right) of the Steingletscher study area. The red rectangle in the map indicates the location of the proglacial study area. In the aerial image, the outlines of the four study moraines are shown. Source: SwissTopo ('Journey through time' - maps and images, year 2019). Modified and reproduced with authorization of the Swiss Federal Office of Topography (2019; BA19029). (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

even during the summer months (MeteoSwiss, 2016a).

The study area is located in the highly polymetamorphic "Erstfelder"-gneiss-zone of the internal alpine massif (Aar massif). The bedrock consists mainly of biotite-plagioclase-gneiss, pre-mesozoic metagranitoids and amphibolites (Labhart, 1977). The glacier forefield is characterized by well-preserved moraine remnants beyond the little-ice age extend of the glacier (King, 1974). Schimmelpfennig et al. (2014) created a detailed and comprehensive chronology of the moraines based on beryllium-10 dating. Based on her work and aerial images and the glacier extent maps for the last 150 years (Swiss Federal Office of Topography - 'Journey through time' - Map and 'SWISSIMAGE Journey through time' - Aerial Images), we selected four moraines that could be dated precisely (Fig. 1b). The two older moraines are ten and two to three thousand years in age and are hereafter referred to as the 10K moraine and 3K moraine. They have a north-west and southern exposition, respectively. The two younger moraines are north-exposed and are approximately 160 and 30 years in age (hereafter called 1860 and 1990 moraine). The younger moraines are located ~600 m south of the older moraines and are 70 m (1990 moraine) and 100 m (1860 moraine) higher in elevation (Fig. 1).

The vegetation cover differed between the moraines and decreased from the 10K moraine (average of 93%; range 90–100%) to the 3K moraine (average: 72%; range: 60–85%) and the 1990 moraine (average: 42%; range: 30–50%); the 1860 moraine did not fit this trend and had a higher vegetation cover (average: 85%; range: 80–95%) than the 3K and 1990 moraine (Table 1) because of the fewer scree fields. The vegetation on the 10 K moraine consists largely of alpine rose (*Rhododendron ferrugineum*), interspersed with nard grass (*Nardus stricta*), common heather (*Calluna vulgaris*) and blueberry (*Vaccinium vulgaris, Vaccinium uliginosum*). These species generate acidic topsoil conditions by root exudates and create a rooty, humus-rich surface (Ellenberg and Leuschner, 2010). The south facing slopes of the 3K moraine are characteristic of dry alpine grasslands and mainly consist of tufted species that form hummocky clusters in between the scree fields. Dwarf carline thistle (*Carlina acaulis*), yarrow (*Achillea moschata*) and stocks of thyme (*Thymus ploytrichus*), mountain cranberry (*Vaccinium vitis-idaea*) and clover (*Trifolium nivale*) are also common. On the 1860 moraine, willow species (*Salix retusa, Salix glaucoserica*) that grow on acid soils in alpine surroundings, and alpine sweet vernalgrass (*Anthoxanthum alpinum*), bellflowers (*Campanula scheuchzeri*) and pale clover (*Trifolium pallescens*) are common. The 1990 moraine is characterized by pioneer plant species. Pale clover (*Trifolium pallescens*), alpine bluegrass (*Poa alpina*) and alpine willowherb (*Epilobium fleischeri*) form scree plant communities, and are colonized by pioneer shrub species (*Salix retusa, Salix hastata*). These are typical early-successional communities on coarse-textured soils and have shallow root systems (Hudek et al., 2017b; Jonasson and Callaghan, 1992; Lichtenegger, 1996; Pohl et al., 2011).

The soils were classified after the World Reference Base for soil resources (IUSS Working Group WRB, 2015) by Musso et al. (2019). The silty-loamy Dystric Cambisol of the 10K moraine has a moderately developed soil and litter layer. The O horizon (0–20 cm depth) is dominated by thick roots and decomposed organic material. The A horizon (20–35 cm) is dark brown to black in color and contains fine roots. The B horizon (35–50 cm) is rich in clay. The saprolite underneath was less weathered and rich in coarse particles (Fig. 2a). The Skeletic Cambisol of the 3K moraine has as sandy loam texture, a thinner organic layer, and less developed horizons than the 10K moraine. The fraction of coarse particles (> 2 mm) was higher and there was no clay-rich B horizon (Fig. 2b). The soils of the 1860 and 1990

Table 1
Overview of the main characteristics of the 12 study plots.

Moraine	Vegetation Complexity	Elevation (m.a.s.l.)	Slope (°)	Aspect (°)	Dominant Vegetation	Vegetation Cover (%)	Soil Type
10K	Low	1882	24	39 NE	*Rhododendron ferrugineum, Vaccinium myrtillus*	100	*Dystric Cambisol*
10K	Medium	1882	29	16 N	*Rhododendron ferrugineum, Vaccinium uliginosum*	90	*Dystric Cambisol*
10K	High	1873	18	43 NE	*Rhododendron ferrugineum, Calluna vulgaris*	90	*Dystric Cambisol*
3K	Low	1914	32	174 S	*Carlina acaulis, Achillea moschata*	60	*Skeletic Cambisol*
3K	Medium	1910	32	169 S	*Vaccinium vitis-idaea, Carlina acaulis*	85	*Skeletic Cambisol*
3K	High	1888	25	141 SE	*Thymus polytrichus, Trifolium nivale*	70	*Skeletic Cambisol*
1860	Low	1989	25	49 NE	*Anthoxanthum alpinum, Salix retusa*	80	*Hyperskeletic Leptosol*
1860	Medium	1981	31	56 NE	*Campanula scheuchzeri, Trifolium pallescens*	80	*Hyperskeletic Leptosol*
1860	High	1989	26	47 NE	*Salix glaucosericea, Anthoxanthum alpinum*	95	*Hyperskeletic Leptosol*
1990	Low	1952	21	48 NE	*Salix hastata*	50	*Hyperskeletic Leptosol*
1990	Medium	1959	34	56 NE	*Epilobium fleischeri Poa alpina*	30	*Hyperskeletic Leptosol*
1990	High	1955	23	28 NE	*Salix retusa, Trifolium pallescens*	45	*Hyperskeletic Leptosol*

Table 2
List of hillslope characteristics and the abbreviations used in the Figures.

	Characteristic	*Abbreviation*	*Unit*
	Moraine Age	*MA*	years
	Saturated Hydraulic Conductivity	K_{sat}	mm hr^{-1}
Topographic	Exposition	*Ex*	Cardinal directions
	Slope	*Sl*	(°)
Surface	Microtopography	*MT*	(–)
	Surface Resistance	*SR*	N cm^{-2}
	Hydrophobicity	*Hy*	(–)
Soil	Clay Content	*ClC*	(%)
	Silt Content	*SiC*	(%)
	Sand Content	*SaC*	(%)
	Gravel Content	*GrC*	(%)
	Macroporosity	*Mp*	(%)
	Soil Organic Matter	*SOM*	(%)
	Soil Aggregate Stability	*SAS*	(–)
Vegetation	Vegetation Complexity	*VCP*	(–)
	Vegetation Cover	*VC*	(%)
	Root Density	*RD*	g cm^{-3}
	Root Length	*RLD*	cm cm^{-3}

moraines were classified as sandy-loamy Hyperskeletic Leptosols (Fig. 2c & d). The organic content and root density is lower than on the older moraines. At the 1990 moraine, we observed a fine layer of rock-flour at a depth of ~ 35 cm. The material was well-sorted, fine-grained and had a blueish hue, suggesting reduced conditions (Fig. 2d).

3. Methods

3.1. Plot selection

We selected three plots (each 4 m by 6 m) on each of the four moraines along the chronosequence and took the same measurements at all 12 plots. The location of the plots on the moraines was chosen based on the structural vegetation complexity (Table 1) in order to cover as much variability within each moraine as possible. Vegetation complexity was assessed in the field for ten vegetation plots (each 1 m by 1 m) on each moraine based on a combination of vegetation cover and functional diversity. The functional diversity was calculated based on eight traits characterizing plants species along the main axes of plant performance (Garnier et al., 2016): specific leaf area, nitrogen content, leaf dry matter content, Raunkiaeŕs life form, seed mass, clonal growth organ, root type and stem growth form. In addition, the number of different species was counted for each vegetation plot. The three chosen plots for the measurements were located around the two vegetation plots with the lowest, intermediate and highest vegetation complexity. Since the vegetation complexity includes the vegetation cover, the plot with the lowest vegetation complexity also represented the plot with the lowest vegetation cover and highest degree of bareness on each moraine.

The exposition of the plots was north-northeast for all the plots, except for the plots on the 3K moraine, which had a southern exposition. The average slope of the plots (determined with an inclinometer) was fairly similar, varying between 23.6° on the 10K moraine to 29.8° on the 3K moraine (Table 1). The microtopography of the plots was determined for ten transects per plot using a 1.5 m long micro-topography profiler consisting of a wooden frame with 101 adjustable 1-m long metal rods (cf. Leatherman, 1987). The normalized measured

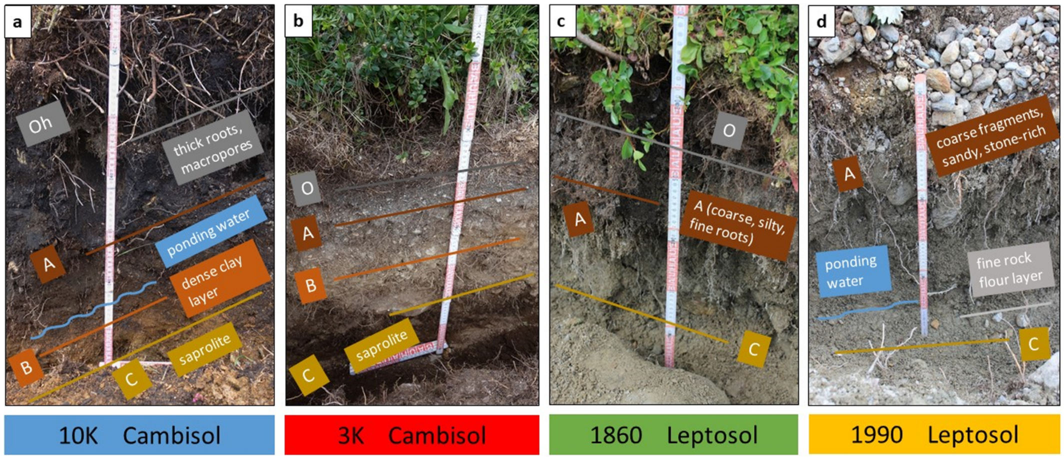


Fig. 2. Photos of the soil profiles of the four moraines (a = 10K moraine, b = 3K moraine, c = 1860 moraine, d = 1990 moraine). Sequences of soil horizons are indicated (grey = organic surface layer, brown = A horizon, light brown = B horizon, yellow = C horizon. The blue lines indicate lower permeability layers on which ponding of water and lateral flow was observed. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

line length was averaged for the ten transects to calculate the Tortuosity Index, as defined by Bertuzzi et al. (1990) and was highest on the 10K moraine (mean Tortuosity Index of 0.95) and lowest on the 1860 moraine (mean Tortuosity Index of 0.31).

3.2. Field measurements

3.2.1. Saturated hydraulic conductivity (K_{sat})

At each plot, K_{sat} was measured at three locations at three different depths (0–5 cm, 5–20 cm and 20–40 cm below the soil surface). For all measurements, the K_{sat} was derived from the steady state infiltration rate. For the surface (0–5 cm), we used a Double-Ring-Infiltrometer with an inner diameter of 20 cm (according to the ASTM D3385-03

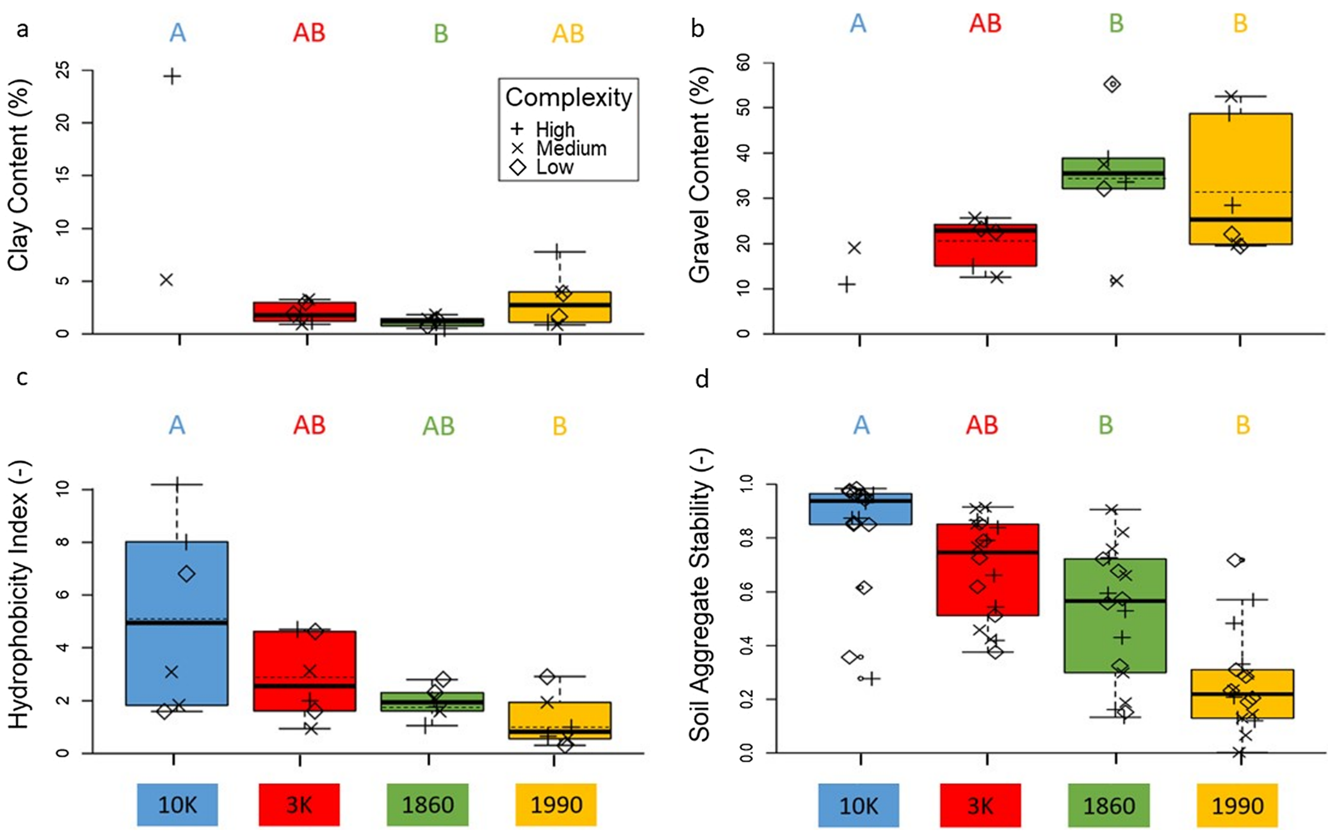


Fig. 3. Boxplots of the clay (a) and gravel (b) contents at 10 cm below the surface, soil hydrophobicity at the surface (c), and soil aggregate stability at 0–10 cm depth for the different moraines. The box extends from the 25th to 75th percentiles and whiskers from the 10th and 90th percentiles; the median is represented by the solid thick line and the average by the dashed line. The jittered symbols represent the actual measurements on the different plots on each moraine. Different capital letters above the graph denote statistically significant different median values for the different moraines. Note that for the 10K moraine, only two soil cores were available for soil texture analysis and thus no box plot is shown.

standard test method). The Double-Ring-Infiltrometer is suitable for the determination of hydraulic properties on sloping landscapes (Bodhinayake et al., 2004) but to facilitate vertical insertion of the infiltrometers into the steep slopes (at least 5 cm), the lower end was cut at angles of 20° and 30°, coresponding to the median and average slope of the moraines. The depth of ponding varied between 15 and 25 cm. The K_{sat} in the subsurface was measured using a Constant Compact Head Permeameter (Amoozometer) (Amoozegar, 1992, 1989). The diameter of the borehole was 6 cm; the height of the water level was 15 cm for the measurements at 5–20 cm depth and 20 cm for the measurements at 20–40 cm depth.

We analyzed the nine measurements per moraine separately, but also report the median, average and standard deviation of K_{sat} per plot. We acknowledge that many more measurements are needed to obtain a robust average value of the K_{sat} per plot (see for example Harden and Scruggs, 2003; Zimmermann, 2008) but this was not possible due to time limitations. However, these limited measurements for multiple plots per moraine already indicate some of the expected variability and trends.

3.2.2. Hydrophobicity

The hydrophobicity of the soil surface was determined at two different locations per plot (below the most dominant species and the second most dominant species) using a Mini Disk Infiltrometer (Decagon Devices, Inc.); the average value was used to represent the hydrophobicity for each plot. The disk infiltrometer was filled with either water or a 95% ethanol solution and infiltration volumes were recorded at a 20-s interval until a steady-flow was achieved. The organic material (lichens, mosses, leafs and litter) was carefully removed with a trowel prior to the measurements. Lichner et al. (2007) proposed an index of soil water repellency, R, based on the work of Tillmann et al. (1989) and Hallett and Young (1999), which is the product of the quotient of the sorptivities (S) of 95% ethanol and water and a factor of 1.95 (Hallett et al., 2001). The index R is unit-less and a proportional estimate of how much sorptivity is reduced by the water repellency of soils. Values of R larger than 1.95 indicate sub-critically water repellent soils (Tillmann et al., 1989). The method is faster and less subjective than other methods, such as the water drop penetration time test (Robichaud et al., 2008).

3.2.3. Surface resistance and soil aggregate stability

Soil surface resistance was measured with an Eijkelkamp hand penetrometer at two different locations per plot (also at the two most dominant vegetation species). The average value was used to represent the resistance for each plot.

For soil aggregate stability analysis, six soil samples were collected from the 0 to 10 cm soil layer per plot using a Humax probe, which consists of a cylindrical stainless steel corer and a 5 kg sledge hammer. Soil aggregate stability was determined for these cores according to the method for stone-rich soils of Bast et al. (2015). This method is based on the wet-sieving method using only a sieve of 20 mm. An aggregate stability coefficient (ASC) of 0 indicates complete dispersion, whereas a value of 1 indicates a completely stable soil sample. The median ASC-value of the six cores was used to represent the aggregate stability for each plot.

3.2.4. Soil texture

At each plot, two soil cores were collected at 10 cm and 30 cm below the surface. The particle size was analyzed using the aerometer method (dry sieving and sedimentation analysis) at the German Research Centre for Geosciences. For the analyses, it was assumed that the texture at 10 cm depth also represents the texture at 0–5 cm depth. We used the average value of the two cores per depth to represent the texture of each plot. For the 10K moraine, only two samples were available for the top 10 cm of the soil layer and these samples were thus assumed to represent the texture for all three plots. We defined particles < 2 µm as clay, 2–63 µm in size as silt, > 63 µm − 2 mm as sand and > 2 mm − 20 mm as gravel.

3.2.5. Soil macroporosity

We used the X-ray computational tomography (XRT) data in Fig. 7 of Musso et al. (2019) to determine the soil macropore content for each moraine. The macropore content was defined as the ratio of the volume of all pores larger than ~ 14 mm^3 (representing spherical macropores with a diameter of 3 mm; cf. Beven and Germann, 1982) and the total volume of pores in the soil core. We used the average value for the two soil cores (0–5 cm depth) per moraine to represent the macropore content for all plots on a moraine.

3.2.6. Soil organic matter

Musso et al. (2019) took samples adjacent to the soil macropore cores from 0 to 20 cm depth to determine the organic matter content by loss on ignition at 550 °C. The average organic matter content of the two soil cores per moraine was assumed to provide an estimate of the soil organic matter content for all plots on each moraine.

3.2.7. Root length density and root density

Root length density and root density were determined in the lab of the Chair of Geobotany at the Faculty of Biology at the University of Freiburg, Germany. In order to analyze the root distribution, six 30 cm long soil cores were taken from each plot. Roots were sorted from intersected depths of 5-, 15- and 25 cm of the soil core using a wet-washing procedure and stored in a freezer until further analysis. The roots were then separated from the remaining soil, prepared for scanning with a flatbed scanner and analyzed using WinRHIZO (Régent Instruments Inc.). The dry weight of the roots was determined after drying for at least 1 week at 60 °C. Root length density and root density were obtained by dividing the root length and the dry root mass by the soil volume, which was corrected for the skeleton content (rock fragments, cobbles, gravel and laterite concretions or ironstones with a diameter > 2 mm). The median values for the six cores was used to represent the root length density and root density for each plot.

3.3. Data analyses

We determined the normality of the data using the Shapiro-Wilk-test. We tested whether the differences in the mean surface hydrological characteristics for the four moraines were statistically significant using one-way ANOVA and Tukey tests for the normally distributed data. If there were more than three values available per moraine (K_{sat}, hydrophobicity, soil aggregate stability, soil surface resistance, soil texture, root density and root length density) or the data were non-normally distributed, we used the Kruskal-Wallis tests and Nemenyi tests to test if the differences in the median values were statistically significant. We used a 0.05 level of significance for all analyses.

To determine the correlations between all measured surface and near surface characteristics, we created a covariance matrix and used the Spearman rank correlation coefficient to determine the strength and direction of the correlations. We used a Random Forest Analysis to determine the dominant factors that affect the surface K_{sat}. Random Forest Analysis is a machine learning algorithm based on multiple decision trees, where each decision tree in the forest considers a random subset of predictors from the dataset. Strong predictors have a higher predictive power (listed on top of the ranking) and have a large impact on the decision trees, so that excluding them increases the mean standard error of the model. We used the highest ranked factors from the Random Forest Analysis in a Multiple Linear Model to predict K_{sat}. We, furthermore, used a Structural Equation Model to highlight the functional chain of the different factors that affect the surface K_{sat}. Structural Equation Models seek to explain the relationships among multiple independent and dependent variables. They examine the structure in a series of equations and answer the question which of the independent variables has a greater effect on the dependent variable in a multiple regression analysis using the standardized regression coefficients (beta factors). These beta factors indicate how many standard deviations the dependent variable changes per standard deviation increase of the predictor variable. Structural Equation Models are particularly useful when the variables have different units. For the Random Forest Analysis, the Multiple Linear Model and the Structural Equation Model, we log-transformed all non-normally-distributed variables (clay content, exposition, hydrophobicity, macropores, moraine age, root density, saturated hydraulic conductivity, silt content, soil surface resistance). For the analyses we used the software R (v3.5.1 – used with RStudio v1.1.463) and in particular the packages: “stats”, “corrplot”, “randomForest”, “lavaan” and “semPlot”.

4. Results

4.1. Soil texture and macroporosity

The texture at 10 cm below the soil surface differed for the four moraines: loam for the 10K moraine, sandy loam for the 3K moraine and 1990 moraine, and loamy sand for the 1860 moraine. The clay content was highest but also most variable for the 10K moraine (5% for one sample and 24% for the other sample). It was lower for the 3K moraine (average: 2%) and 1860 moraine (average: 1%) and slightly higher for the 1990 moraine (average: 3%) (Fig. 3a). The silt content of the top 10 cm of the soil varied more than the clay content (averaging at 30%, 14%, 5%, and 8% for the 10K, 3K, 1860 and 1990 moraine, respectively). The average sand content was highest for the 10K moraine (40%) and lowest for the 3K moraine (20%), and intermediate for the 1860 and 1990 moraines (30% and 28%, respectively). The gravel content was lowest for the 10K moraine (average: 15%), intermediate for the 3K moraine (average: 21%) and highest but also most variable on the 1860 moraine (average: 35%, but varying between 12% and 55%) and the 1990 moraine (average: 32%; varying between 19% and 53%) (Fig. 3b). Gravel, sand, silt and clay contents at 30 cm depth did not differ significantly from the surface layer, although the differences for the clay and gravel factions between the moraines were less pronounced than at 10 cm depth. Surprisingly, soil macroporosity was highest on the 1860 moraine (34% and 42%) and lowest on the 1990 moraine (12% and 38%).

4.2. Root length density and root density, soil organic matter and soil aggregate stability

Median root length density at 5 cm below the surface was almost the same for the 10K, 3K and 1860 moraine (0.41 cm cm^{-3}, 0.44 cm cm^{-3}, 0.43 cm cm^{-3}, respectively) but was much lower for the youngest moraine (0.25 cm cm^{-3}). Root length densities decreased with depth for all the moraines, except the 1990 moraine, where root length densities dropped from 0.25 cm cm^{-3} at 5 cm to 0.18 cm cm^{-3} at 15 cm and 25 cm depth. Root density near the surface (5 cm depth) decreased along the chronosequence (median values of 0.15 g cm^{-3}, 0.09 g cm^{-3}, 0.06 g cm^{-3}, and 0.03 g cm^{-3} for the plots on the 10K, 3K, 1860 and 1990 moraines, respectively). The same trend was observed for the deeper soil layers (15 cm and 25 cm depth). At all plots, except on the 10K moraine, the root density decreased with depth below the surface. At the 10K moraine the root density was highest at 15 cm depth (average of the median values of the three plots of 0.15 g cm^{-3}, 0.29 g cm^{-3} and 0.06 g cm^{-3} at 5, 15 and 25 cm respectively).

Soil organic matter was highest on the moraine (38% and 44%), lowest on the 1990 moraine (0.8% and 1.1%), and intermediate at the 3K (3% and 22%) and 1860 moraine (4% and 38%). Soil aggregate stability decreased from the 10K moraine to the 1990 moraine (median aggregate stability coefficients of 0.94, 0.75, 0.57 and 0.22 for the 10K, 3K, 1860 and 1990 moraine, respectively; Fig. 3d). The median aggregate stability coefficients for the two oldest moraines were significantly different from the median aggregate stability coefficients for the two youngest moraines. None of these soil characteristics were related to vegetation complexity (Fig. 3).

4.3. Hydrophobicity and soil surface resistance

Hydrophobicity decreased from the 10K to the 1990 moraine (10K moraine: R = 4.9, 3K moraine: R = 2.6, 1860 moraine: R = 1.9, and 1990 moraine: R = 0.8; Fig. 3c). However, only the difference between the median values of the 10K moraine and the 1990 moraine was statistically significant. This was in part due to the very large range in hydrophobicity for the 10K moraine (R = 1.6–10.2). Thus some parts of the soil on the oldest moraines were sub-critically water repellent but the soil on the youngest moraine was not.

The median soil surface resistance was significantly different between the moraines (p = 0.02) and increased from the 10K moraine (median = 90 N cm^{-2}) to the 3K moraine (median: 130 N cm^{-2}) and the 1860 moraine (median: 145 N cm^{-2}), but dropped again on the 1990 moraine (median: 90 N cm^{-2}).

4.4. Saturated hydraulic conductivity

The saturated hydraulic conductivity (K_{sat}) decreased quickly with depth below the surface for all moraines (Fig. 4). The median saturated hydraulic conductivity (K_{sat}) increased along the chronosequence from the 10K moraine to the 1990 moraine, except that K_{sat} at 20–40 cm depth for the 1990 moraine was lower than for the 3K and 1860 moraine (Fig. 4). The variability in the K_{sat} values was largest for the 10K moraine and smallest for the 3K moraine (coefficient of variation of the log-transformed K_{sat} data of 0.63 and 0.33, respectively) (Fig. 4).

4.5. Correlations between K_{sat} and soil characteristics

4.5.1. K_{sat} and root density

There was a significant negative linear correlation between the median surface saturated hydraulic conductivity (K_{sat}) and median root length density and root density at 5 cm for the plots (Fig. 5a and d). The lower K_{sat} values of for the moraine coincided with a higher root length density and root density, whereas for the youngest moraine K_{sat} values

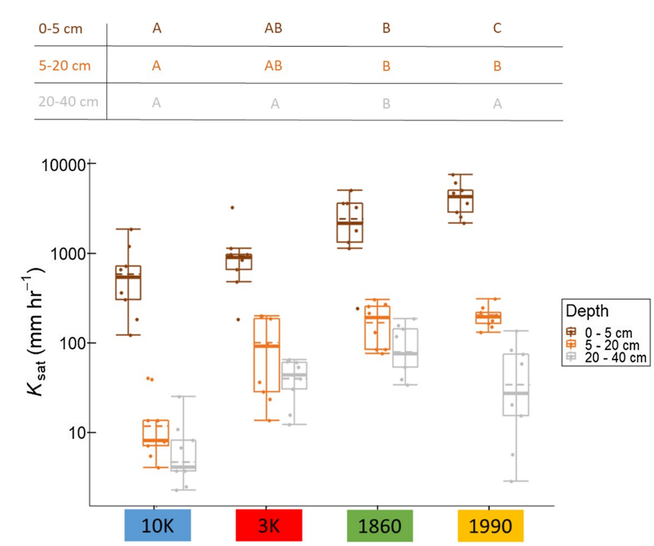


Fig. 4. Box plot of the K_{sat} values for the three depths (0–5 cm, dark brown; 5–20 cm, light brown; 20–40 cm, grey) for the four moraines. Different capital letters above the figure denote statistically significant differences in the median K_{sat} values between the four moraines. See the caption of Fig. 3 for the description of the box plots. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

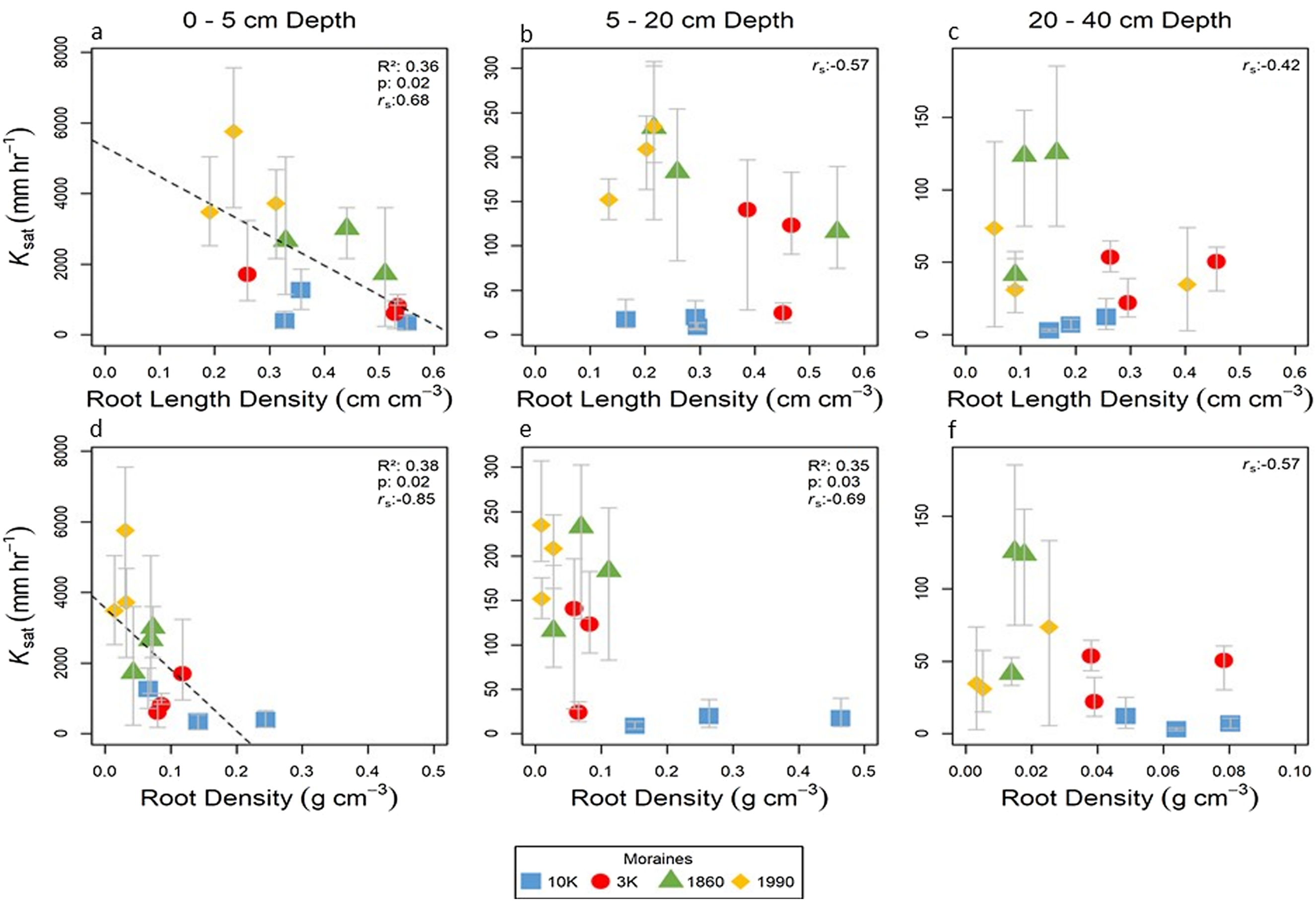


Fig. 5. Scatterplots showing the relation between the median K_{sat} at the surface (left column), 5–20 cm depth (middle), and 20–40 cm depth (right) and the median root length density (top) and median root density (bottom) for each plot. Note that measurements of the surface K_{sat} are plotted against root length density and root density at 5 cm depth (a, d), K_{sat} measurements between 5 and 20 cm depth are plotted against root length density and root density at 15 cm depth (b, e), and K_{sat} measurements between 20 and 40 cm depth are plotted against root length density and root density measured at 25 cm depth. The colors indicate the individual moraines; each symbol represents one plot. The error bars extend to the highest and lowest K_{sat} value measured on each plot. Statistically significant linear relationships are highlighted by the dashed line. The p-value and coefficient of determination R^2, as well as the Spearman rank correlation coefficient (r_s) are shown in the upper right corner of each subplot. Note that the x-axis of subplot f differs from the x-axis of plots d and e.

were highest and roots were the least abundant and shortest. There was no significant correlation between K_{sat} and root length density for the two deepest soil layers (5–20 cm and 20–40 cm; Fig. 5b–c). There was a statistically significant negative correlation between K_{sat} and root density at 20 cm (Fig. 5e) but this was largely due to the data from the 10 K moraine forming a separate cluster. There was no significant correlation between K_{sat} and root density at 20–40 cm (Fig. 5f).

4.5.2. K_{sat} and soil texture

There was a negative correlation between K_{sat} and the silt content and a positive correlation between K_{sat} and gravel for the surface (Fig. 6d, j) and at 5–20 cm depth (Fig. 6e, k). There was no statistically significant correlation between the clay content and K_{sat} for the top soil layer (Fig. 6a). There was a statistically significant correlation between the clay content and K_{sat} at 5–20 cm (Fig. 6b) but this relation was highly influenced by the high clay content for the 10K moraine. There was no correlation between K_{sat} and the sand content (Fig. 6g, h, i) because the sand content did not differ significantly between the different moraines. There were no statistically significant corelations between K_{sat} at 20–40 cm depth and soil texture (Fig. 6c, f, i, l).

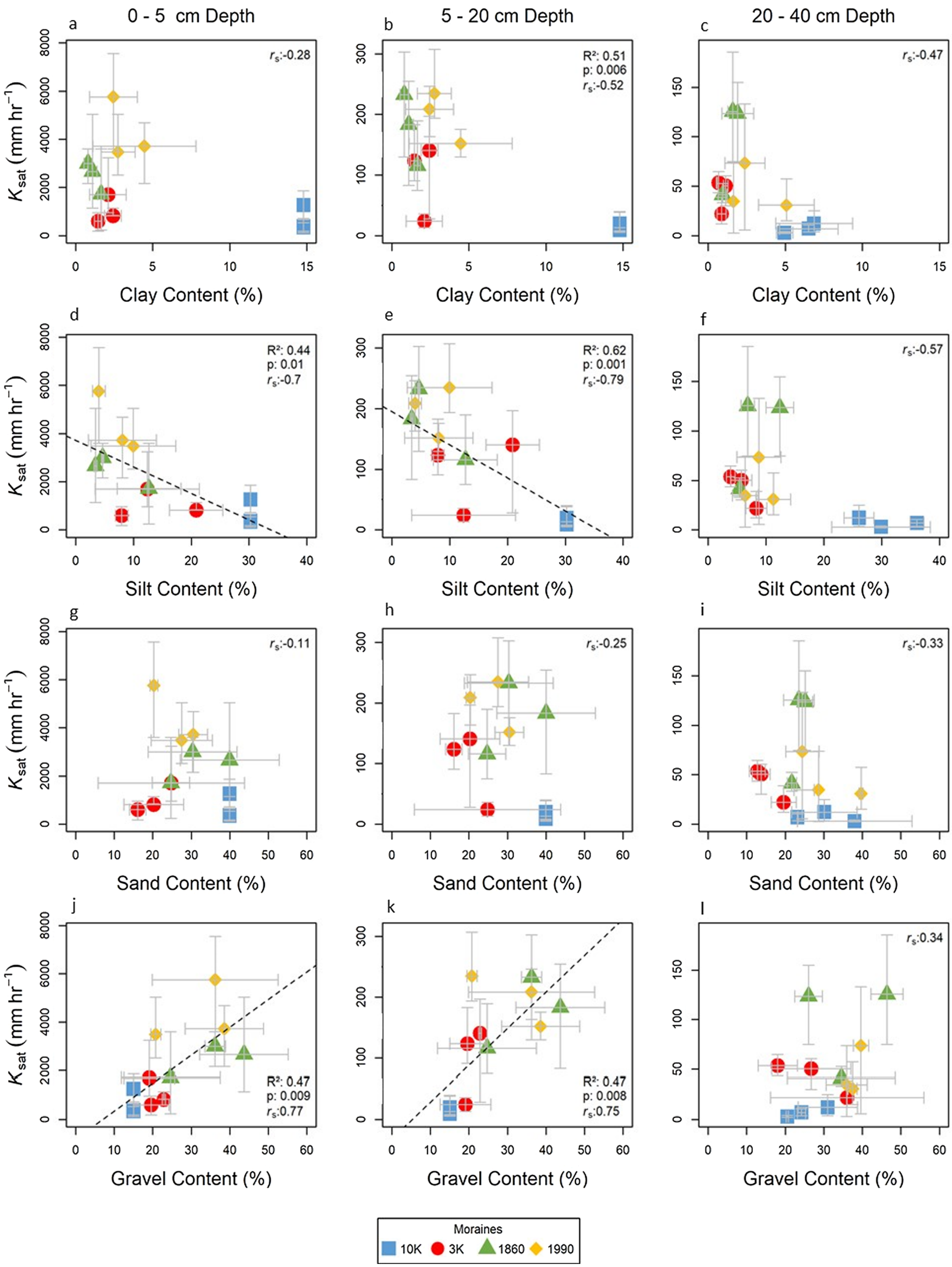
4.5.3. Surface K_{sat} and other surface characteristics

The correlations between the surface K_{sat} and the site characteristics were analyzed in further detail because most other site characteristics were measured at the surface and the patterns in K_{sat} for the different depths were similar to those at the surface (Fig. 4). The covariance matrix summarizes the corelations between all investigated hillslope characteristics (Fig. 7) and suggests that K_{sat} was mainly a function of vegetation traits, soil properties and the hillslope age. K_{sat} was negatively correlated with the silt content (SiC, $r_s = -0.70$; Fig. 6d), the soil organic matter (SOM, $r_s = -0.69$; Fig. 8i), vegetation cover (VC, $r_s = -0.61$; Fig. 8d), root length density and root density (RLD, r_s = -0.68 and RD, $r_s = -0.85$), as well as soil hydrophobicity (Hy, r_s = -0.94; Fig. 8k), soil aggregate stability (SAS, $r_s = -0.87$; Fig. 8j) and moraine age (MA, $r_s = -0.93$; Fig. 4), with the strongest correlations for hydrophobicity and moraine age. The gravel content (GrC, $r_s = 0.77$) was the only site characteristic that was positively correlated with the surface K_{sat} (as already shown in Fig. 6j and k). There was no correlation between K_{sat} and any of the topographic characteristics (Fig. 8a-c). K_{sat} was also not correlated to the soil suface resistance (Fig. 8g), macroporosity (Fig. 8h), vegetation complexity (Fig. 8e) or the number of species (Fig. 8f).

4.6. Random Forest Analysis

The Random Forest Analysis reveals the relative importance of the individual hillslope characteristics to explain the variability in the median K_{sat}. The ranking (Fig. 9) suggests that K_{sat} can be partly explained by a combination of pedological factors (hydrophobicity, gravel content and soil aggregate stability) and the moraine age, and that these factors have a larger effect on K_{sat} than vegetation related characteristics (vegetation cover, root length density, and root density). However, it has to be kept in mind that the ranking of these soil parameters in the Random Forest Analysis expresses the predictive power of an independent variable on K_{sat}, but doesn't quantify its impact.

The Multiple Linear Model based on the top two ranked site characteristics from the Random Forest Analysis (hydrophobicity Hy and



(caption on next page)

Fig. 6. Scatterplots of the relation between the median K_{sat} at the surface (left), 5–20 cm depth (middle) and 20–40 cm depth (right) and the average clay (top), silt (second row), sand (third row) and gravel content (bottom) for each plot. Note that K_{sat} at the surface and at 5–20 cm depth is plotted against the texture at 10 cm depth; K_{sat} at 20–40 cm depth is plotted against the texture at 30 cm depth. Only two soil samples at 10 cm depth were available for texture analysis for the 10K moraine and the same average value is thus used for all three plots. The color of the symbols represents the different moraines; each symbol represents one plot. The error bars extend to the highest and lowest K_{sat} value measured on each plot (vertical bar) and to the highest and lowest particle content for each plot (horizontal bar). Statistically significant linear relationships are highlighted by the dashed line; the p-value and coefficient of determination R^2, and the Spearman rank correlation coefficients (r_s) are given in the upper left or right corner of each subplot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

gravel content GrC) had only a slightly higher coefficient of determination (log K_{sat} = 5.94–1.03Hy + 0.76GrC; R^2 = 0.82; Root mean Squared Error RMSE = 0.34 mm h^{-1}) than the correlation with moraine age MA (log K_{sat} = 9.65–0.36MA; R^2 = 0.77; Root mean Squared Error RMSE = 0.41 mm h^{-1}).

4.7. Structural Equation Model

The relations between the most important biotic and abiotic hillslope characteristics for K_{sat} can be summarized in a Structural Equation Model (Fig. 10). It clearly shows the large number of interactions or interdependencies between soil (left) and vegetation (right). Particularly soil aggregate stability (SAS) and hydrophobicity (Hy) are soil characteristics that are highly affected by vegetation. The beta factors that describe the strength of the connections between two attributes, highlight the importance of the gravel content (GrC) as one of the major factors that controls the variation in K_{sat}.

5. Discussion

5.1. Changes in vegetation and soil characteristics with moraine age

As expected vegetation cover increased with moraine age but the 3K moraine had a lower vegetation cover than expected based on this trend. This was probably caused by the plot selection: there were small patches of scree on two of the three plots on the 3K moraine, reducing the plant covered area. The trend in vegetation cover explains the variation in soil organic matter content (Fig. 8i), root length (Fig. 5a) and root density (Fig. 5d). However, the number of species did not change systematically with slope age (Fig. 8f). The plots on the 3K moraine had the largest number of species (Fig. 8f). Perhaps this is related to the southern exposition of the slope; solar radiation affects plant diversity and the degree of colonization on temperate mountains (Winkler et al., 2016).

The increase of the silt content (Fig. 6d, e) and the decrease in the gravel content (Fig. 6j, k) with moraine age reflects the soil development by physical and chemical weathering. Physical weathering leads to a decrease in grain size with increasing soil age (D'Amico et al., 2014; Egli et al., 2001; Mavris et al., 2010) and is at higher elevations mainly due to comminution of soil particles by congelifraction (Oztas and Fayetorbay, 2003). Chemical weathering, i.e., the leaching of primary minerals and the formation of secondary minerals, such as clay minerals, is accelerated by the growth of vegetation and the release of root exudates that cause a breakdown of the primary soil minerals. Despite the large variation in clay content within the moraines (Fig. 6c) and the limited number of samples for the moraine, the trend of clay formation along the chronosequence is a sign of advanced soil development on the oldest moraine. The high variability in clay content for the two samples from the 10K moraine is likely due to differences in the wetness conditions across the slopes as soil weathering rates are higher

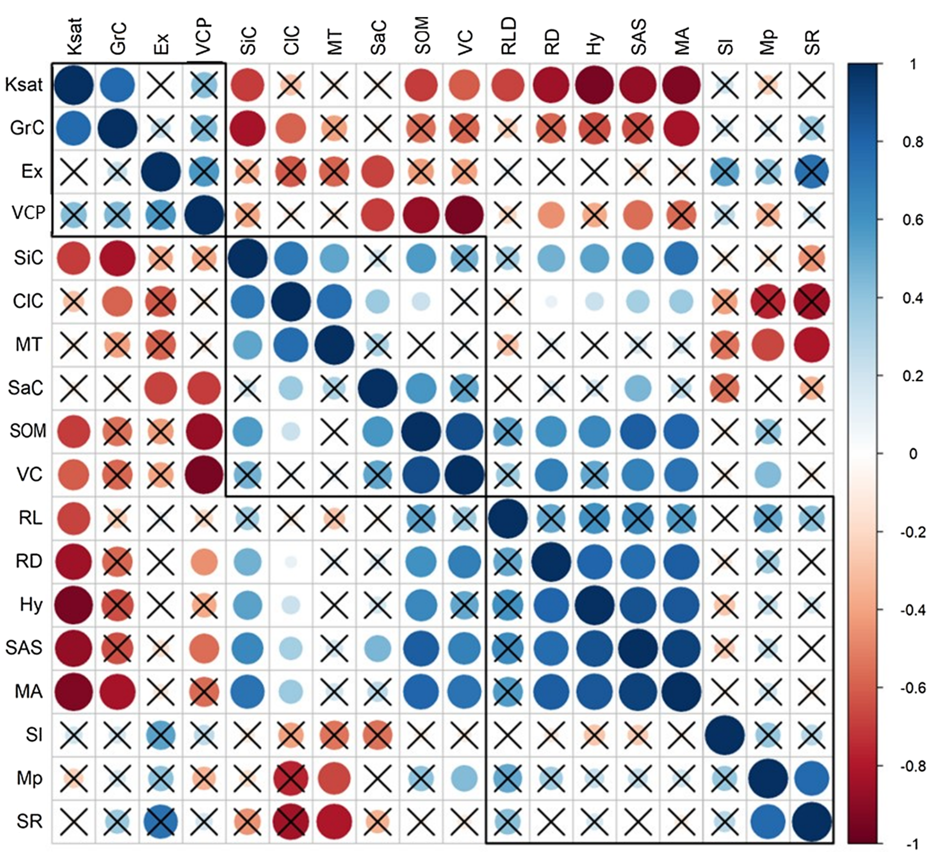


Fig. 7. Covariance matrix of the surface characteristics for the plots along the moraine chronosequence in the proglacial area of the Steingletscher. The size of the circle and intensity of the color reflect the strength of the Spearman rank correlation. Crosses represent non-significant relationships. Black rectangles separate hierarchical clusters of the covariance matrix. For the abbreviations of the characteristics, see Table 2.

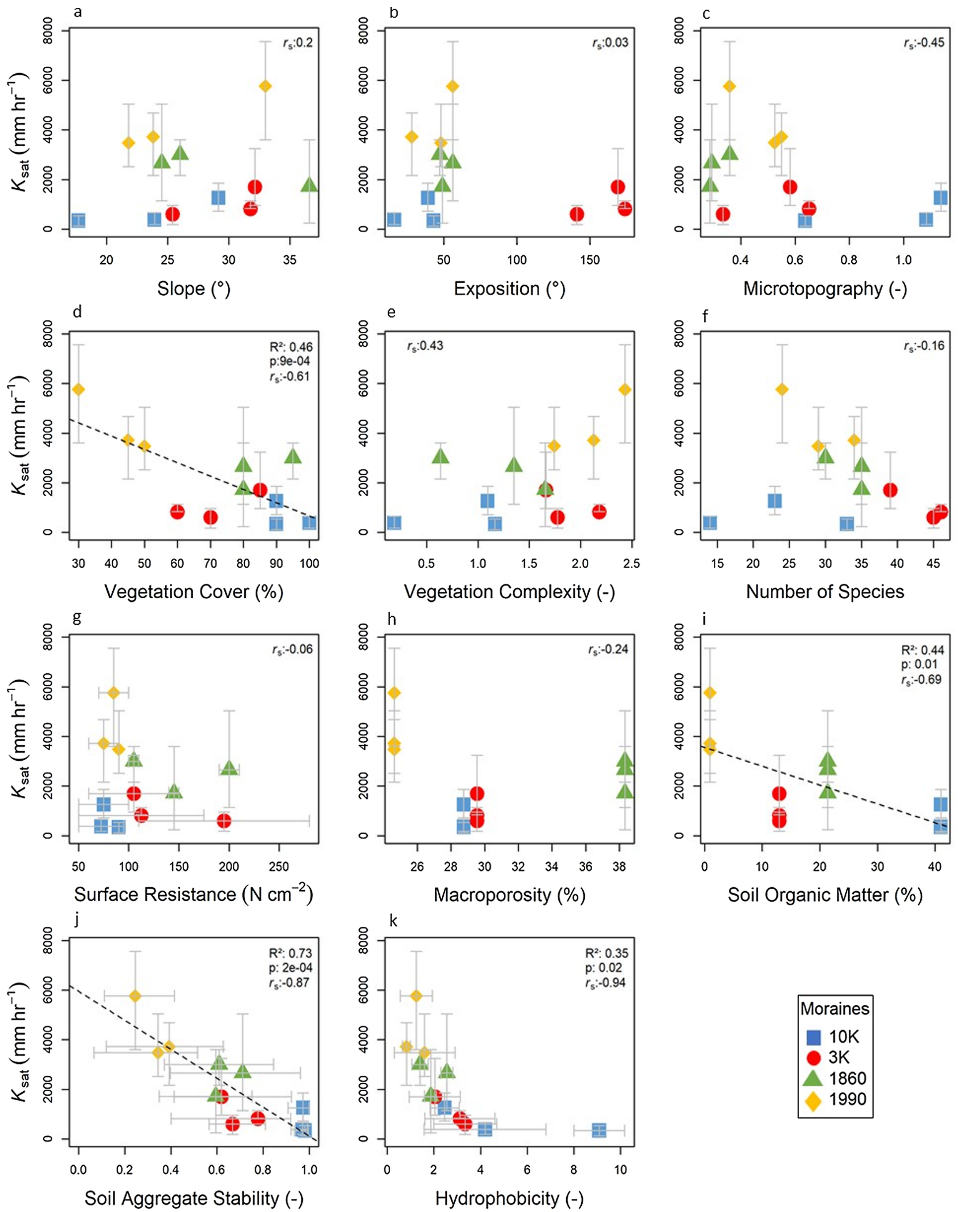


Fig. 8. Scatterplots of the relation between the median K_{sat} at the surface and the hillslope surface characteristics. The values for the surface resistance and hydrophobicity represent the average value per plot. Note that the macroporosity was only determined for two cores per moraine and that thus the same average value is plotted for all three plots per moraine. The colors indicate the different moraines; each symbol represents a plot. The grey error bars extend from the lowest to highest measured value for each plot. Statistically significant linear relationships are highlighted by the dashed line; the p-value, coefficient of determination R^2 and Spearman rank correlation coefficients (r_s) are given in the upper left or right corner of each subplot. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

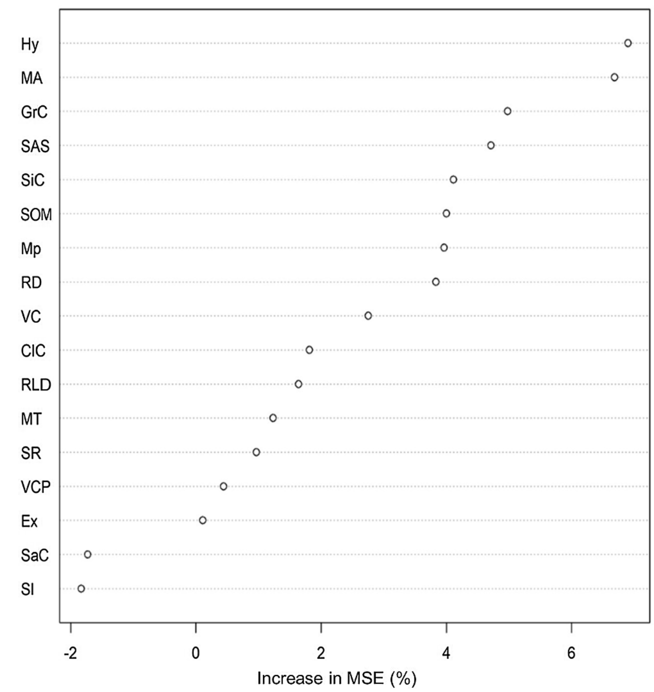


Fig. 9. Predictive power of the site characteristics for the surface saturated hydraulic conductivity (K_{sat}) as represented by the increase in the mean standard error (MSE, %) when omitting the variable from the Random Forest Analysis. See Table 2 for the abbreviations of the site characteristics.

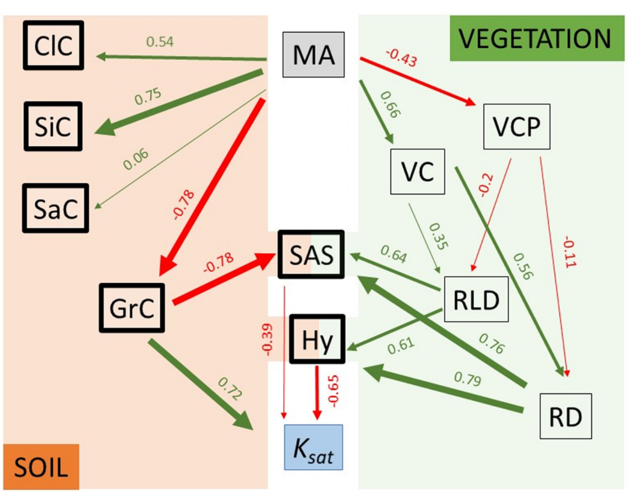


Fig. 10. The Structural Equation Model of the interrelationships between the vegetation and soil characteristics for the study plots on the moraine chronosequence of the Steingletscher forefield. The median saturated hydraulic conductivity K_{sat} is the dependent variable at the end of the functional chain of biological (right side) and pedological (left side) characteristics. Hillslope age is the origin of the functional chain and drives the evolution of both soil and vegetation characteristics and therefore also K_{sat}. The values represent the normalized regression coefficients (beta factors). Green arrows indicate positive correlations, red colors negative correlations. The thickness of the arrow represents the strength of the correlation. The thickness of the outline of the boxes signifies the importance of these soil characteristics for K_{sat}. For the abbreviations of the site characteristics see Table 2. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

in areas of higher soil moisture (Egli et al., 2015).

Temporal variation in soil aggregate stability is caused by the activity of fungal communities (Duchicela et al., 2013), or the formation of fine root systems and small particles during soil development (Erktan et al., 2016). The higher soil organic matter content for the 10K moraine likely resulted in more stable soil aggregates by reducing slaking and decreasing the wettability of the aggregates (Sullivan, 1990). Furthermore, the roots on the older moraines might have increased soil aggregate stability by physical enmeshment of particles and root mucilage (Morel et al., 1991). In addition to the differences in vegetation and thus roots, also the changes in the particle size distribution with moraine age might have contributed to the differences in soil aggregate stability. On the 10K moraine, binding forces of clay minerals likely stabilized the soil particles (Lado et al., 2004), whereas on the 1990 moraine, coarse gravel fragments cause poor aggregation. However, the deeper fine textured rock flour layer on the 1990 moraine also had stable soil aggregates (Fig. 2d).

Variations in hydrophobicity are typically attributed to seasonal changes in climate and soil moisture, impacts of wildfires, or water-repellent plant exudates (Buczko et al., 2006; Ferreira et al., 2000; Jiménez-Pinilla et al., 2016). On the 10K moraine, the dense shrubs of alpine rose (*Rhododendron ferrugineum)* possibly increased soil hydrophobicity (Fig. 3a) due to the acidic humus-rich soil surface (Fig. 2a) and because the roots release water-repellent exudates (Ellenberg and Leuschner, 2010). Other studies have also reported higher values of the *R*-Index in forest and shrubland than grasslands and meadows (Lichner et al., 2007). Lichner (2007) reported *R*-Index-values ranging from about 0.7 for shallow grassland soils up to about 210 for pine forest soils (although the median value for the 10K plots was only 4.9 and the maximum R-value was 10.2). Since values above 1.95 have a negative effect on the sorptivity (Tillmann et al., 1989), hydrophobicity might have slightly decreased infiltration rates on the 10K moraine.

5.2. Comparison of the measured K_{sat} values to the literature

The measured K_{sat}-values were generally very high, especially for the surface, where the median K_{sat} was higher than 500 mm hr^{-1} for all moraines and maximum values were larger than 4000 mm hr^{-1}. These are typical conductivities for sand and gravel (Hendriks, 2010). The high K_{sat} values for the Steingletscher moraines are comparable to the K_{sat} values reported for the few other K_{sat} studies on young moraines. King et al. (2019) conducted comprehensive measurements of surface K_{sat} of different surface units in the proglacial area of the Eklutina watershed in southcentral Alaska. The moraines in the catchment consist of till and outwash deposits, ranging in age from late-Pleistocene to a couple of decades. The reported average K_{sat}-values varied between ~460 mm hr^{-1} to ~4300 mm hr^{-1} on outwash deposits and talus cones that have a comparable grain size distribution as the moraines in the proglacial area of the Steingletscher. Barontini et al. (2005) measured surface infiltrations on moraines in the Italian Toce River basin in the Central Alps. For the forest-covered moraines, they measured K_{sat} values up to ~1200 mm hr^{-1}, which compares reasonably well to the maximum K_{sat} value measured on the 10K moraine (1260 mm hr^{-1}). Their surface infiltration rates up to 2900 mm hr^{-1} for grassland moraines are comparable to the K_{sat} values for the 1860 moraine in this study (up to 3000 mm hr^{-1}; median 2660 mm hr^{-1}). MacDonald et al. (2012) reported K_{sat} -values up to ~ 4000 mm hr^{-1} for very loose, sandy and gravelly glacial moraine soils in northeast Scotland, which agree well with the surface infiltration rates on the 1990 moraine (up to 5760 mm hr^{-1}; average: 4320 mm hr^{-1}).

The decrease in K_{sat}-values with moraine age (Fig. 4) is in agreement with findings of other chronosequence studies. Topsoil K_{sat} decreased from ~ 83 mm hr^{-1} to ~ 4 mm hr^{-1} in desert soils of 50 and 100,000 years of age, respectively (Young et al., 2004). For the subsurface (20–50 cm depth), K_{sat} decreased from 140 mm hr^{-1} to 0.8 mm hr^{-1} on volcanic soils of 300 and 4 million years of age, respectively (Lohse and Dietrich, 2005). Both studies also found a significant relationships between K_{sat} and the soil texture (particularly gravel content). This relation was also found for the Chicken Creek Catchment in Germany (Schaaf et al., 2017).

5.3. Drivers of changes in K_{sat} during hillslope evolution

The overall effect of hillslope age on K_{sat} is the result of the combined effects of the multiple changes in hillslope characteristics. The Structural Equation Model (Fig. 10) can be seen as a theoretical framework of the interactions and feedback processes between the changes in soil and vegetation characteristics in the first 10,000 years of hillslope development. It conceptually outlines moraine age as the starting point and driving force for pedological and biological changes on the moraines and incorporates K_{sat} at the end as the product of the combined effects of these components. The Structural Equation Model indicates that during the first 10,000 years of hillslope development on moraines the effect of changes in soil texture on K_{sat} are larger than the effects of vegetation on K_{sat}. If vegetation was the driving factor of K_{sat}, the hydraulic conductivity would increase with hillslope age. Instead we observed a decrease in K_{sat} with hillslope age (Fig. 4).

The scatterplots suggest a significant negative correlation between root length density, root density and K_{sat} (Fig. 5a, d). This is counterintuitive as K_{sat} and vegetation are typically positive correlated due to flow through soil macropores and preferential flow paths (Alaoui et al., 2011; Ghestem et al., 2011; Hu et al., 2016). These negative relations between K_{sat} and the roots are likely a spurious correlation (rather than a functional relationship) due to the larger effect of the changes in soil texture on K_{sat} as soils develop. This negative relation is opposite to that found in most studies on K_{sat} that show that K_{sat} tends to increase with soil organic matter content (e.g. Eneje et al., 2007; Felton and Ali, 1992; Papadopoulos et al., 2006). The negative correlation between K_{sat} and soil aggregate stability suggests that the high gravel content on the younger moraines decreased soil aggregate stability (Fig. 3d) but didn't reduce the saturated hydraulic conductivity (Fig. 8j) as reported by previous studies (e.g. Abu-Sharar and Salameh, 1995). This is most likely because the particles were too big to clog the flow conducting soil pores. The hydrophobic coatings on the soil surface of the 3K and especially on the 10K moraine (Fig. 7k) due to soil organic matter might have contributed to lower infiltration rates (cf. Nemes et al., 2005) but the hydrophobicity is low and thus can't functionally explain the large variation in K_{sat}, even though it is one of the most highly correlated parameters.

Our results do not suggest that vegetation is not important for the change in K_{sat} during landscape development. The effect of vegetation on K_{sat} likely depends on the period of landscape development. During the initial stages of landscape development (i.e., during the first 10,000 years after deglaciation in this study), the effects of the changes in soil texture on K_{sat} likely outweigh the effects of changes in vegetation as a major driver of changes in K_{sat}. Other chronosequence studies that included young soils suggested that soil texture was the driving factor for the changes in K_{sat} as well (Young et al., 2004; Lohse & Dietrich, 2005). However, these chronosequence studies all included very young soils (only a few centuries or even decades). Thus, even though these studies suggest that during initial landscape development K_{sat} decreases, vegetation might change more than soil texture at later stages of landscape evolution and have a large effect on K_{sat}, as reflected the change from a 'geo-hydro-system towards a bio-geo-hydro system' reported by Biber et al. (2013). This is particularly likely in harsh conditions, such as the alpine areas, where vegetation growth is slow but changes in soil texture may be rapid due to physical weathering. At later stages of hillslope development, larger shrubs or even trees may cause larger macropores and enhance the saturated hydraulic conductivity. An indication for this increase in K_{sat} after initial decreases during landscape evolution is given by the study of Greenwood and Buttle (2014). They report average surface K_{sat} -values of up to $\sim$ 6800 mm hr^{-1} for forest-covered moraines, which are even higher than the measured values for the 1990 moraine. They also found a general increase in K_{sat} from open fields to forests, even though soil texture was comparable at all sites. However, clay layers due to clay illuviation and podzolic layers tend to have a very low K_{sat}, thus K_{sat} may continue to decrease over time, at some depth below the soil surface.

5.4. Implications for runoff generation processes

The value of the near surface K_{sat} relative to the rainfall intensity determines whether rainfall infiltrates into a soil or flows along the surface (Archer et al., 2013; Zimmermann et al., 2013). The depth where K_{sat} becomes less than the maximum or common rainfall intensity determines the likelihood and frequency of saturated overland flow. For the Grimsel Hospiz long term climate station in the Steingletscher region, the 1-hour and 10-min rainfall intensities with a 100-year return interval are 39 mm hr^{-1} (95% confidence interval: 28–62 mm hr^{-1}) and 121 mm hr^{-1} (95% confidence interval: 84–209 mm hr^{-1}), respectively (MeteoSwiss, 2016b). The values for the 2-year return period are 13 mm hr^{-1} (95% confidence interval: 12–14 mm hr^{-1}) and 34 mm hr^{-1} (95% confidence interval: 29–40 mm hr^{-1}), respectively (MeteoSwiss, 2016b). Thus for all moraines, the measured median surface K_{sat}-values are larger than the extreme rainfall intensities. This suggests that infiltration-excess overland flow is unlikely to occur. However, similar to many other locations, K_{sat} decreased sharply with soil depth on all moraines (Fig. 4). At the 10K moraine, the K_{sat} at 20–40 cm below the soil surface was less than 10 mm hr^{-1} (Fig. 4) and thus will likely hamper percolation deeper into the soil during intense events (Fig. 2a). Previous studies have also suggested that landscape evolution leads to the lower permeability soil layers that hamper water movement and promote lateral subsurface flow (Lohse and Dietrich, 2005; Young et al., 2004). Vertical water movement is likely also restricted on the 1990 moraine, where the K_{sat} of the fine-grained rock flour layer was lower than the high rainfall intensities (Fig. 2d). This suggests that during large high intensity events with high antecedent soil moisture conditions saturation-excess overland flow and potentially also surface erosion may occur on both the youngest and the oldest moraines (Bryan, 2000; Fitzjohn et al., 1998). However, the variability of the measured K_{sat} values within the individual moraines also suggest that there are areas with higher K_{sat}, particularly on the 10K and 3K moraine, where overland flow may re-infiltrate.

The pore size distribution data from Musso et al. (2019) suggested the lowest macroporosity for the 1990 moraine and the highest macroporosity for the 1860 moraine. These results are likely affected by the small size of the cores and the small number of cores per moraine. Blue dye experiments of Semenova (2019) suggested that infiltration at the 10K and 3K moraine was dominated by preferential flow, while the soils on the younger two moraines were conducive to homogeneous matrix flow. Local pipeflow may be generated along the root systems and macropores (Semenova, 2019) in the 3K and 10K moraines and may exfiltrate, and also cause overland flow. During the K_{sat}-measurements on the 10K moraine water was observed to exfiltrate underneath large boulders or patches of alpine rose several meters below the measurement point. Lind and Lundin (1990) reported high-velocities for water flow through macropores next to fine-grained impermeable soil layers for vegetated soils on glacial till in Scandinavia, as well.

In a landscape consisting of moraines of different ages, knowledge of the relation between K_{sat} and moraine age might help to better simulate the runoff responses across the drainage basin. The evolution of K_{sat} in slopes, at least during initial landscape development, might also be important for landscape evolution models or hydrological models that are run for long times. The general trend of decreasing K_{sat} along the chronosequence suggests a higher susceptibility for overland flow during initial landscape development.

5.5. Limitations of the study

Even though the limitations of the space-for-time approach are well-known (Pickett, 1989), we often have to use it because we simply can't take repeated measurements over a long time period (in our case 10,000 years). We had to assume that the moraines were comparable at the time that the glacier retreated (e.g., that the lower gravel content on the 10K moraine was not due to an inherent difference between the moraines). Because the moraines were formed by the same glacier and the trends were relatively clear (i.e., there wasn't one moraine that was always an outlier), this appears to be a reasonable assumption but potential differences cannot be ruled out. However, the rock flour layer wasn’t observed at any of the other moraines and may be a particular feature of the 1990 moraine. We further assumed that the moraines experienced the same climate (or at least it varied similarly) because of the close proximity of the moraines.

To assess whether the sites have an identical (though not equally long) history is impossible over the 10,000 year time scale. We do not know of any disturbances in the past for the younger and older moraines. All moraines might occassionally be affected by low intensity grazing by sheep or cows. However, we assume that this low intensity grazing did not have a significant effect on the K_{sat} values.

We acknowledge that the result may be affected by the relatively small number of measurements for some parameters of the hillslope characteristics. We chose the most diverse plots on each moraine based on above ground vegetation to include as much as possible of the potential variation within each moraine in our measurements. Even though the small sample size for some hillslope characteristics may affect the range, as well as the mean or median values, the trends for K_{sat} and many other parameters are clear and we expect these to hold, even if more data would be available.

6. Conclusions

The near-surface saturated hydraulic conductivity (K_{sat}) is an important hydrologic characteristic because it determines whether precipitation infiltrates into the soil or runs off the surface as overland flow. It also determines the near surface partitioning of precipitation into lateral subsurface flow and vertical drainage. It is, therefore, a crucial factor to describe runoff and erosion, which are two main processes that determine how landscapes change over time. Our study on a moraine chronosequence in a proglacial area in the Central Alps of Switzerland shows that even though vegetation cover, root networks and soil organic matter increase over time, the saturated hydraulic conductivity decreases due to the decline in gravel and increase in silt and clay content. Furthermore, the vegetation may cause some hydrophobicity which may reduce surface infiltration rates. On all moraines, K_{sat} systematically decreased with soil depth but the decline with soil depth was most pronounced for the oldest moraine. The formation of a clay rich layer at 20–40 cm below the soil surface at the oldest moraine results in a lower permeability at this depth and may impede vertical infiltration and enhance lateral flow. This may also imply that the older slopes in the chronosequence are more prone to saturation excess overland flow. These results are important for understanding the spatial variability in runoff generation processes across alpine areas with moraines of different ages, as well as understanding long-term changes in runoff generation processes, and therefore may help to improve hydrological or landscape evolution models.

Acknowledgements

This research is part of the HILLSCAPE project (www.hillscape.ch), which is supported by the Swiss National Science Foundation (project grant number 200021E-167563/1) and the German Research Foundation (DFG project number 318089487). We want to thank Thomas Michel and his team of the Alpin Center Sustenpass for the provision of tools, accommodation and meals, and their logistical support with the fieldwork. We also want to thank Peter Luchs for technical support and permission for conducting fieldwork on his property, and the Kraftwerke Oberhasli AG for the permission to conduct fieldwork on their land. Thanks also to Krzysztof Lamorski (Institute of Agrophysics, Polish Academy of Sciences) for the analysis on soil macropores. Finally we want to thank all of the MSc. students, BSc. Students, interns and staff involved in the project for the help in the field and in the laboratory.

Declaration of Competing Interest

The authors declare no competing interests.

References

Abu-Sharar, T.M., Salameh, A.S., 1995. Reductions in hydraulic conductivity and infiltration rate in relation to aggregate stability and irrigation water turbidity. Agric. Water Manag. 29, 53–62.

Alaoui, A., Caduff, U., Gerke, H.H., Weingartner, R., 2011. Preferential flow effects on infiltration and runoff in grassland and forest soils. Vadose Zo. J. 10, 367. https://doi.org/10.2136/vzj2010.0076.

Alegre, J.C., Cassel, D.K., 1996. Dynamics of soil physical properties under alternative systems to slash-and-burn. Agric. Ecosyst. Environ. 58, 39–48. https://doi.org/10.1016/0167-8809(95)00654-0.

Amoozegar, A., 1992. Advances in measurement of soil physical properties: Bringing theory into practice, Compact constant head permeameter: a convenient device for measuring hydraulic conductivity. Soil Sci. Soc. Am. https://doi.org/10.2136/sssaspecpub30.c3.

Amoozegar, A., 1989. A compact constant-head permeameter for measuring saturated hydraulic conductivity of the vadose zone. Soil Sci. Soc. Am. J. 53 (5), 1356–1361. https://doi.org/10.2136/sssaj1989.03615995005300050009x.

Archer, N.A.L., Bonell, M., Coles, N., MacDonald, A.M., Auton, C.A., Stevenson, R., 2013. Soil characteristics and landcover relationships on soil hydraulic conductivity at a hillslope scale: a view towards local flood management. J. Hydrol. 497, 208–222. https://doi.org/10.1016/j.jhydrol.2013.05.043.

Barontini, S., Clerici, A., Ranzi, R., Bacchi, B., 2005. Saturated hydraulic conductivity and water retention relationships for Alpine mountain soils. In: Climate and Hydrology of Mountain Areas, pp. 101–121.

Bast, A., Wilcke, W., Graf, F., Lüscher, P., Gärtner, H., 2015. A simplified and rapid technique to determine an aggregate stability coefficient in coarse grained soils. Catena 127, 170–176. https://doi.org/10.1016/j.catena.2014.11.017.

Bedford, D.R., Small, E.E., 2008. Spatial patterns of ecohydrologic properties on a hillslope-alluvial fan transect, central New Mexico. Catena 73, 34–48. https://doi.org/10.1016/j.catena.2007.08.005.

Bernasconi, S.M., Bauder, A., Bourdon, B., Brunner, I., Bünemann, E., Chris, I., Derungs, N., Edwards, P., Farinotti, D., Frey, B., Frossard, E., Furrer, G., Gierga, M., Göransson, H., Gülland, K., Hagedorn, F., Hajdas, I., Hindshaw, R., Ivy-Ochs, S., Jansa, J., Jonas, T., Kiczka, M., Kretzschmar, R., Lemarchand, E., Luster, J., Magnusson, J., Mitchell, E.A.D., Venterink, H.O., Plötze, M., Reynolds, B., Smittenberg, R.H., Stähli, M., Tamburini, F., Tipper, E.T., Wacker, L., Welc, M., Wiederhold, J.G., Zeyer, J., Zimmermann, S., Zumsteg, A., 2011. Chemical and biological gradients along the Damma Glacier Soil Chronosequence, Switzerland. Vadose Zo. J. 10, 867. https://doi.org/10.2136/vzj2010.0129.

Bertuzzi, P., Rauws, G., Courault, D., 1990. Testing roughness indices to estimate soil surface roughness changes due to simulated rainfall. Soil Tillage Res. 17, 87–99. https://doi.org/10.1016/0167-1987(90)90008-2.

Beven, K., 2018. A century of denial: preferential and nonequilibrium water flow in soils, 1864–1984. Vadoze Zo. J. 17, 17. https://doi.org/10.2136/vzj2018.08.0153.

Beven, K., Germann, P., 2013. Macropores and water flow in soils revisited. Water Resour. Res. 49, 3071–3092. https://doi.org/10.1002/wrcr.20156.

Beven, K., Germann, P., 1982. Macropores and water flow in soils. Water Resour. Res. 18, 1311–1325. https://doi.org/10.1029/WR018i005p01311.

Biber, P., Seifert, S., Zaplata, M.K., Schaaf, W., Pretzsch, H., Fischer, A., 2013. Relationships between substrate, surface characteristics, and vegetation in an initial ecosystem. Biogeosciences 10, 8283–8303. https://doi.org/10.5194/bg-10-8283-2013.

Bodhinayake, W., Bing Cheng, S., Kosuke, N., 2004. Determination of hydraulic properties in sloping landscapes from tension and double-ring infiltrometers. Vadose Zo. J. 3 (3), 964–970. https://doi.org/10.2113/3.3.964.

Bormann, H., Klaassen, K., 2008. Seasonal and land use dependent variability of soil hydraulic and soil hydrological properties of two Northern German soils. Geoderma 145, 295–302. https://doi.org/10.1016/j.geoderma.2008.03.017.

Bryan, R.B., 2000. Soil erodibility and processes of water erosion on hillslope. Geomorphology 32, 385–415. https://doi.org/10.1016/S0169-555X(99)00105-1.

Buczko, U., Bens, O., Hüttl, R.F., 2006. Water infiltration and hydrophobicity in forest soils of a pine – beech transformation chronosequence. J. Hydrol. 331, 383–395. https://doi.org/10.1016/j.jhydrol.2006.05.023.

Bundt, M., Widmer, F., Pesaro, M., Zeyer, J., Blaser, P., 2001. Preferential flow paths: biological “hot spots” in soils. Soil Biol. Biochem. 33, 729–738. https://doi.org/10.1016/S0038-0717(00)00218-2.

Burga, C.A., Krüsi, B., Egli, M., Wernli, M., Elsener, S., Ziefle, M., Fischer, T., Mavris, C., 2010. Plant succession and soil development on the foreland of the Morteratsch glacier (Pontresina, Switzerland): straight forward or chaotic? Flora-Morphology. Distrib. Funct. Ecol. Plants 205, 561–576. https://doi.org/10.1016/j.flora.2009.10.001.

Cammeraat, L.H., Imeson, A.C., 1999. The evolution and significance of soil-vegetation patterns following land abandonment and fire in Spain. Catena 37, 107–127. https://doi.org/10.1016/S0341-8162(98)00072-1.

Cerdà, A., 1997. Seasonal changes of the infiltration rates in a Mediterranean scrubland on limestone. J. Hydrol. 198 (1–4), 209–225. https://doi.org/10.1016/S0022-1694(96)03295-7.

Clark, E.V., Zipper, C.E., 2016. Vegetation influences near-surface hydrological characteristics on a surface coal mine in eastern USA. Catena 139, 241–249. https://doi.org/10.1016/j.catena.2016.01.004.

D'Amico, M.E., Freppaz, M., Filippa, G., Zanini, E., 2014. Vegetation influence on soil formation rate in a proglacial chronosequence (Lys Glacier, NW Italian Alps). Catena 113, 122–137. https://doi.org/10.1016/j.catena.2013.10.001.

D'Amico, M.E.D., Freppaz, M., Leonelli, G., Bonifacio, E., Zanini, E., 2015. Early stages of soil development on serpentinite: the proglacial area of the Verra Grande Glacier, Western Italian Alps. J. Soils Sediments 15, 1292–1310. https://doi.org/10.1007/s11368-014-0893-5.

Duchicela, J., Sullivan, T.S., Bontti, E., Bever, J.D., 2013. Soil aggregate stability increase is strongly related to fungal community succession along an abandoned agricultural field chronosequence in the Bolivian Altiplano. J. Appl. Ecol. 50, 1266–1273. https://doi.org/10.1111/1365-2664.12130.

Dunkerley, D., 2000. Hydrologic effects of dryland shrubs: defining the spatial extent of modified soil water uptake rates at an Australian desert site. J. Arid Environ. 45, 159–172. https://doi.org/10.1006/jare.2000.0636.

Egli, M., Lessovaia, S.N., Chistyakov, K., 2015. Microclimate affects soil chemical and mineralogical properties of cold alpine soils of the Altai Mountains (Russia). J. Soils Sedim. 15, 1420–1436. https://doi.org/10.1007/s11368-013-0838-4.

Egli, M., Mavris, C., Mirabella, A., Giaccai, D., 2010. Soil organic matter formation along a chronosequence in the Morteratsch proglacial area (Upper Engadine, Switzerland). Catena 82, 61–69. https://doi.org/10.1016/j.catena.2010.05.001.

Egli, M., Mirabella, A., Fitze, P., 2001. Clay mineral formation in soils of two different chronosequences in the Swiss Alps. Geoderma 104, 145–175. https://doi.org/10.1016/S0016-7061(01)00079-9.

Elhakeem, M., Papanicolaou, A.N.T., Wilson, C.G., Chang, Y., Burras, L., Abban, B., Wysocki, D.A., Wills, S., 2018. Understanding saturated hydraulic conductivity under seasonal changes in climate and land use. Geoderma 315, 75–87. https://doi.org/10.1016/j.geoderma.2017.11.011.

Ellenberg, H., Leuschner, C., 2010. Vegetation Mitteleuropas mit den Alpen: in ökologischer, dynamischer und historischer Sicht, Volume 810. ed. Utb.

Eneje, R.C., Ogbenna, C.V., Nuga, B.O., 2007. Saturate hydraulic conductivity, water stable aggregates and soil organic matter in a sandy-loam soil in Ikwuano lga of Abia state. Agro-Science 4, 34–37. https://doi.org/10.4314/as.v4i1.1519.

Eppes, M.C., Bierma, R., Vinson, D., Pazzaglia, F., 2008. A soil chronosequence study of the Reno valley, Italy: insights into the relative role of climate versus anthropogenic forcing on hillslope processes during the mid-Holocene. Geoderma 147, 97–107. https://doi.org/10.1016/j.geoderma.2008.07.011.

Erktan, A., Cécillon, L., Graf, F., Roumet, C., Legout, C., Rey, F., 2016. Increase in soil aggregate stability along a Mediterranean successional gradient in severely eroded gully bed ecosystems: combined effects of soil, root traits and plant community characteristics. Plant Soil 398, 121–137. https://doi.org/10.1007/s11104-015-2647-6.

Evans, D.J.A., 1999. A soil chronosequence from neoglacial moraines in western norway. Geogr. Ann. Ser. A, Phys. Geogr. 81 (1), 47–62. https://doi.org/10.1111/j.0435-3676.1999.00048.x.

Evans, D.M., Zipper, C.E., Hester, E.T., Schoenholtz, S.H., 2015. Hydrologic effects of surface coal mining in Appalachia (U.S.). J. Am. Water Resour. Assoc. 51, 1436–1452. https://doi.org/10.1111/1752-1688.12322.

Felton, G.K., Ali, M., 1992. Hydraulic parameter response to incorporated organic matter in the B-horizon. Trans. ASAE 35, 1153–1160. https://doi.org/10.13031/2013.28713.

Ferreira, A.J.D., Coelho, C.O.A., Walsh, R.P.D., Shakesby, R.A., Ceballos, A., Doerr, S.H., 2000. Hydrological implications of soil water-repellency in Eucalyptus globulus forests, north-central Portugal. J. Hydrol. 231–232, 165–177. https://doi.org/10.1016/S0022-1694(00)00192-X.

Fischer, C., Tischer, J., Roscher, C., Eisenhauer, N., Ravenek, J., Gleixner, G., Attinger, S., Jensen, B., de Kroon, H., Mommer, L., Scheu, S., Hildebrandt, A., 2015. Plant species diversity affects infiltration capacity in an experimental grassland through changes in soil properties. Plant Soil 397, 1–16. https://doi.org/10.1007/s11104-014-2373-5.

Fitzjohn, C., Ternan, J.L., Williams, A.G., 1998. Soil moisture variability in a semi-arid gully catchment: implications for runoff and erosion control. Catena 32, 55–70. https://doi.org/10.1016/S0341-8162(97)00045-3.

Garnier, E., Navas, M.L., Grigulis, K., 2016. Plant Functional Diversity: Organism Traits, Community Structure, and Ecosystem Properties. Oxford University Press.

Germer, S., Neill, C., Krusche, A.V., Elsenbeer, H., 2010. Influence of land-use change on near-surface hydrological processes: undisturbed forest to pasture. J. Hydrol. 380, 473–480. https://doi.org/10.1016/j.jhydrol.2009.11.022.

Gerwin, W., Schaaf, W., Biemelt, D., Fischer, A., Winter, S., Hüttl, R.F., 2009. The artificial catchment "Chicken Creek" (Lusatia, Germany)—A landscape laboratory for interdisciplinary studies of initial ecosystem development. Ecol. Eng. 35, 1786–1796. https://doi.org/10.1016/j.ecoleng.2009.09.003.

Ghestem, M., Sidle, R.C., Stokes, A., 2011. The influence of plant root systems on subsurface flow: implications for slope stability. Bioscience 61, 869–879. https://doi.org/10.1525/bio.2011.61.11.6.

Ghimire, C.P., Bonell, M., Bruijnzeel, L.A., Coles, N.A., Lubczynski, M.W., 2013. Reforesting severely degraded grassland in the Lesser Himalaya of Nepal: effects on soil hydraulic conductivity and overland flow production. J. Geophys. Res. Earth Surf. 118, 2528–2545. https://doi.org/10.1002/2013JF002888.

Greenwood, W.J., Buttle, J.M., 2014. Effects of reforestation on near-surface saturated hydraulic conductivity in a managed forest landscape, southern Ontario, Canada. Ecohydrology 7, 45–55. https://doi.org/10.1002/eco.1320.

Hallett, P.D., Baumgartl, T., Young, I.M., 2001. Subcritical water repellency of aggregates from a range of soil management practices. Soil Biol. Biochem. 65, 435–439. https://doi.org/10.1037/a0029756.

Hallett, P.D., Young, I.M., 1999. Changes to water repellence of soil aggregates caused by substrate-induced microbial activity. Eur. J. Soil Sci. 50, 35–40. https://doi.org/10.1046/j.1365-2389.1999.00214.x.

Harden, C.P., Scruggs, P.D., 2003. Infiltration on mountain slopes: a comparison of three environments. Geomorphology 55, 5–24. https://doi.org/10.1016/S0169-555X(03)00129-6.

Helm, D.J., Allen, E.B., Helm, D.J., 1995. Vegetation Chronosequence near Exit Glacier, Kenai Fjords National Park, Alaska, U.S.A. Arct. Alp. Res. 27, 246–257. https://doi.org/10.1080/00040851.1995.12003119.

Hendriks, M.R., 2010. Introduction to Physical Hydrology. Oxford University Press.

Hirmas, D.R., Giménez, D., Nemes, A., Kerry, R., Brunsell, N.A., Wilson, C.J., 2018. Climate-induced changes in continental-scale soil macroporosity may intensify water cycle. Nature 561, 100. https://doi.org/10.1038/s41586-018-0463-x.

Holländer, H.M., Blume, T., Bormann, H., Buytaert, W., Chirico, G.B., Exbrayat, J.-F., Gustafsson, D., Hölzel, H., Kraft, P., Stamm, C., Stoll, S., Blöschl, G., Flühler, H., 2009. Comparative predictions of discharge from an artificial catchment (Chicken Creek) using sparse data. Hydrol. Earth Syst. Sci. 13, 2069–2094. https://doi.org/10.5194/hess-13-2069-2009.

Hu, X., Li, Z.C., Li, X.Y., Liu, L.Y., 2016. Quantification of soil macropores under alpine vegetation using computed tomography in the Qinghai Lake Watershed, NE Qinghai-Tibet Plateau. Geoderma 264, 244–251. https://doi.org/10.1016/j.geoderma.2015.11.001.

Huang, C., Gascuel-odoux, C., Cros-Cayot, S., 2001. Hillslope topographic and hydrologic effects on overland flow and erosion. Catena 46, 177–188. https://doi.org/10.1016/S0341-8162(01)00165-5.

Hudek, Stanchi, S., D'Amico, M., Freppaz, M., 2017a. Quantifying the contribution of the root system of alpine vegetation in the soil aggregate stability of moraine. Int. Soil Water Conserv. Res., 5, 36–42. https://doi.org/10.1016/j.iswcr.2017.02.001.

Hudek, Sturrock, C.J., Atkinson, B.S., Stanchi, S., Freppaz, M., 2017b. Root morphology and biomechanical characteristics of high altitude alpine plant species and their potential application in soil stabilization. Ecol. Eng., 109, 228–239. https://doi.org/10.1016/j.ecoleng.2017.05.048.

IUSS Working Group WRB, 2015. World Reference Base for Soil Resources 2014, update 2015. World International soil classification system for naming soils and creating legends for soil maps. World Soil Resources Reports No. 106. FAO, Rome.

Jarvis, N., Koestel, J., Messing, I., Moeys, J., Lindahl, A., 2013. Influence of soil, land use and climatic factors on the hydraulic conductivity of soil. Hydrol. Earth Syst. Sci. 17, 5185–5195. https://doi.org/10.5194/hess-17-5185-2013.

Jefferson, A., Grant, G.E., Lewis, S.L., Lancaster, S.T., 2010. Coevolution of hydrology and topography on a basalt landscape in the Oregon Cascade. Earth Surf. Process. Landforms 816, 803–816. https://doi.org/10.1002/esp.1976.

Jiménez-Pinilla, P., Lozano, E., Mataix-Solera, J., Arcenegui, V., Jordán, A., Zavala, L.M., 2016. Temporal changes in soil water repellency after a forest fire in a Mediterranean calcareous soil: influence of ash and different vegetation type. Sci. Total Environ. 572, 1252–1260. https://doi.org/10.1016/j.scitotenv.2015.09.121.

Johnson, B.G., Layzell, A.L., Cary, M., 2015. Chronosequence development and soil variability from a variety of sub-alpine, post-glacial landforms and deposits in the southeastern San Juan Mountains of Colorado. Catena 127, 222–239. https://doi.org/10.1016/j.catena.2014.12.026.

Johnson, M.S., Lehmann, J., 2016. Double-funneling of trees: stemflow and root-induced preferential flow. Écoscience 13, 324–333. https://doi.org/10.2980/i1195-6860-13-3-324.1.

Jonasson, S., Callaghan, T.V., 1992. Root mechanical properties related to disturbed and stressed habitats in the Arctic. New Phytol. 122, 179–186. https://doi.org/10.1111/j.1469-8137.1992.tb00064.x.

Kelly, E.F., Chadwick, O.A., Hilinski, T.E., 1998. The effect of plants on mineral weathering. Biogeochemistry 42 (1–2), 21–53. https://doi.org/10.1023/A:1005919306687.

King, A.L., Meyers, J.S., Brown, J.M., 2019. Hydraulic conductivity of quaternary surficial units within the Eklutna Valley, southcentral Alaska. J. Hydrol. 575, 166–174. https://doi.org/10.1016/j.jhydrol.2019.05.024.

King, L., 1974. Studien zur postglazialen Vegetationsgeschichte des Sustenpassgebietes. Baseler Beitr. Geogr. 18, 1–123.

Labhart, T.P., 1977. Aarmassiv und Gotthardmassiv. Sammlung Geol. Führer 63, 173.

Lado, M., Ben-Hur, M., Shainberg, I., 2004. Soil wetting and texture effects on aggregate stability, seal formation, and erosion. Soil Sci. Soc. Am. J. 68, 1992–1999. https://doi.org/10.2136/sssaj2004.1992.

Leatherman, S.P., 1987. Field measurement of microtopography. J. Coast. Res. 3, 233–235.

Lichner, Ľ., Hallett, P.D., Feeney, D.S., Ďugová, O., Šír, M., Tesař, M., 2007. Field measurement of soil water repellency and its impact on water flow under different vegetation. Biologia (Bratisl). 62, 537–541. https://doi.org/10.2478/s11756-007-0106-4.

Lichtenegger, E., 1996. Root distribution in some alpine plants. Acta Phytogeogr. Suec. 81, 76–82.

Lind, B.B., Lundin, L., 1990. Saturated hydraulic conductivity of Scandinavian Tills. Nord. Hydrol. 21, 107–118.

Liu, Y., Miao, H.T., Chang, X., Wu, G.L., 2019. Higher species diversity improves soil water infiltration capacity by increasing soil organic matter content in semiarid grasslands. L. Degrad. Dev. 1599–1606. https://doi.org/10.1002/ldr.3349.

Lohse, K.A., Dietrich, W.E., 2005. Contrasting effects of soil development on hydrological properties and flow paths. Water Resour. Res. 41, 1–17. https://doi.org/10.1029/2004WR003403.

Lyford, F.P., Qashu, H.K., 1969. Infiltration rates as affected by desert vegetation. Water Resour. Res. 5, 1373–1376. https://doi.org/10.1029/WR005i006p01373.

MacDonald, A.M., Maurice, L., Dobbs, M.R., Reeves, H.J., Auton, C.A., 2012. Relating in situ hydraulic conductivity, particle size and relative density of superficial deposits in a heterogeneous catchment. J. Hydrol. 434–435, 130–141. https://doi.org/10.1016/j.jhydrol.2012.01.018.

Mao, J., Nierop, K.G.J., Rietkerk, M., Sinninghe, J.S., Dekker, S.C., 2016. The influence of vegetation on soil water repellency-markers and soil hydrophobicity. Sci. Total Environ. 566–567, 608–620. https://doi.org/10.1016/j.scitotenv.2016.05.077.

Martinez, L.J., Zinck, J.A., 2004. Temporal variation of soil compaction and deterioration of soil quality in pasture areas of Colombian Amazonia. Soil Till. Res. 75, 3–17. https://doi.org/10.1016/j.still.2002.12.001.

Mavris, C., Egli, M., Plötze, M., Blum, J.D., Mirabella, A., Giaccai, D., Haeberli, W., 2010. Initial stages of weathering and soil formation in the Morteratsch proglacial area (Upper Engadine, Switzerland). Geoderma 155, 359–371. https://doi.org/10.1016/j.geoderma.2009.12.019.

Mayor, Á.G., Bautista, S., Bellot, J., 2009. Factors and interactions controlling infiltration, runoff, and soil loss at the microscale in a patchy Mediterranean semiarid landscape. Earth Surf. Process. Landforms 34 (12), 1702–1711. https://doi.org/10.1002/esp.

Mbonimpa, M., Aubertin, M., Chapuis, R.P., Bussie, B., 2002. Practical pedotransfer functions for estimating the saturated hydraulic conductivity. Geotech. Geol. Eng. 235–259. https://doi.org/10.1023/A:1016046214724.

MeteoSwiss, 2016a. Swiss Federal Office of Meteorology and Climatology. Swiss climate in detail. Climate diagrams. Norm values tables. [WWW Document]. URL < https://www.meteoswiss.admin.ch/product/output/climate-data/climate-diagrams-normal-values-station-processing/GRH/climsheet_GRH_np8110_e.pdf > (accessed 1.10.18).

MeteoSwiss, 2016b. Swiss Federal Office of Meteorology and Climatology. Swiss climate in detail. Extreme value analysis. Standard period 1966 - 2015. [WWW Document]. URL < https://www.meteoschweiz.admin.ch/home/klima/schweizer-klima-im-detail/extremwertanalysen/standardperiode.html?station = sap + (20180613) > (accessed 1.10.18).

Morel, J.L., Habib, L., Plantureux, S., Guckert, A., 1991. Influence of maize root mucilage on soil aggregate stability. Plant Soil 136, 111–119. https://doi.org/10.1007/BF02465226.

Moreno-de las Heras, M., Merino-Martin, L., Nicolau, J.M., 2009. Effect of vegetation cover on the hydrology of reclaimed mining soils under Mediterranean-Continental climate. Catena 77, 39–47. https://doi.org/10.1016/j.catena.2008.12.005.

Mori, A.S., Osono, Æ.T., Uchida, Æ.M., 2008. Changes in the structure and heterogeneity of vegetation and microsite environments with the chronosequence of primary succession on a glacier foreland in Ellesmere Island, high arctic Canada. Ecol. Res. 23, 363–370. https://doi.org/10.1007/s11284-007-0388-6.

Musso, A., Lamorski, K., Slawinski, C., Geitner, C., Hunt, A., Greinwald, K., 2019. Evolution of soil pores and their characteristics in a siliceous and calcareous proglacial area. Catena 182. https://doi.org/10.1016/j.catena.2019.104154.

Nemes, A., Rawls, W.J., Pachepsky, Y.A., 2005. Influence of organic matter on the estimation of saturated hydraulic conductivity. Soil Sci. Soc. Am. J. 69, 1330–1337. https://doi.org/10.2136/sssaj2004.0055.

Nicolau, J.M., Puigdefdbregas, J., Gutidrrez, L., 1996. Effects of soil and vegetation on runoff along a catena in semi-arid Spain. Geomorphology 14, 297–309. https://doi.org/10.1016/0169-555X(95)00043-5.

Oztas, T., Fayetorbay, F., 2003. Effect of freezing and thawing processes on soil aggregate stability. Catena 52, 1–8. https://doi.org/10.1016/S0341-8162(02)00177-7.

Papadopoulos, A., Bird, N.R.A., Whitmore, A.P., Mooney, S.J., 2006. The effects of organic farming on the soil physical environment. Asp. Appl. Biol. 79, 5.

Papanicolaou, A., Elhakeem, M., Wilson, C.G., Burras, C.L., West, L.T., Lin, H., Clark, B., Oneal, B.E., 2015. Spatial variability of saturated hydraulic conductivity at the hillslope scale: understanding the role of land management and erosional effect. Geoderma 243–244, 58–68. https://doi.org/10.1016/j.geoderma.2014.12.010.

Pellegrini, S., García, G., Peñas-castejon, J.M., Vignozzi, N., Costantini, E.A.C., 2016. Pedogenesis in mine tails affects macroporosity, hydrological properties, and pollutant flow. Catena 136, 3–16. https://doi.org/10.1016/j.catena.2015.07.027.

Pickett, S.T.A., 1989. Space-for-Time Substitution as an Alternative to Long-Term Studies. In: Likens, G.E. (Ed.), Long-Term Studies in Ecology. Springer Verlag, New York, pp. 110–135.

Pohl, M., Stroude, R., Buttler, A., Rixen, C., 2011. Functional traits and root morphology of alpine plants. Ann. Bot. 108, 537–545. https://doi.org/10.1093/aob/mcr169.

Prieksat, M.A., Kaspar, T.C., Ankeny, M.D., 1994. Positional and temporal changes in ponded infiltration in a corn field. Soil Sci. Soc. Am. J. 58, 181–184. https://doi.org/10.2136/sssaj1994.03615995005800010026x.

Rasmussen, C., Mcguire, L., Dhakal, P., Pelletier, J.D., 2017. Catena Coevolution of soil and topography across a semiarid cinder cone chronosequence. Catena 156, 338–352. https://doi.org/10.1016/j.catena.2017.04.025.

Robichaud, P.R., Lewis, S.A., Ashmun, L.E., 2008. New procedure for sampling infiltration to assess post-fire soil water repellency. US Department of Agriculture, Forest Service, Rocky Mountain Research Station. 14 p., 33. Fort Collins, Colorado. https://doi.org/10.2737/RMRS-RN-33.

Robinson, D.A., Hopmans, J.W., Filipovic, V., Ploeg, M. Van Der, Lebron, I., Jones, S.B., Reinsch, S., Jarvis, N., Tuller, M., 2019. Global environmental changes impact soil hydraulic functions through biophysical feedbacks. Glob. Chang. Biol., pp. 1895–1904. https://doi.org/10.1111/gcb.14626.

Rudner, M., 2005. Environmental patterns and plant communities of the ephemeral wetland vegetation in two areas of the Southwestern Iberian Peninsula. Phytocoenologia 35, 231–265. https://doi.org/10.1127/0340-269X/2005/0035-0231.

Ruggenthaler, R., Meißl, G., Geitner, C., Leitinger, G., Endstrasser, N., Schöberl, F., 2016. Investigating the impact of initial soil moisture conditions on total infiltration by using an adapted double-ring infiltrometer. Hydrol. Sci. J. 61, 1263–1279. https://doi.org/10.1080/02626667.2015.1031758.

Schaaf, W., Elmer, M., Fischer, A., Gerwin, W., Nenov, R., Pretzsch, H., Seifert, S., Winter, S., Zaplata, M.K., 2013. Monitoring the formation of structures and patterns during initial development of an artificial catchment. Environ. Monit. Assess. 185, 5965–5986. https://doi.org/10.1007/s10661-012-2998-x.

Schaaf, W., Pohle, I., Maurer, T., Gerwin, W., Hinz, C., Badorreck, A., 2017. Water Balance Dynamics during Ten Years of Ecological Development at Chicken Creek Catchment. Vadoze Zo. J. 16, 14. https://doi.org/10.2136/vzj2017.04.0074.

Schaap, M.G., Leij, F.J., Van Genuchten, M.T., 2001. Rosetta: a computer program for estimating soil hydraulic parameters with hierarchical pedotransfer functions. J. Hydrol. 251, 163–176. https://doi.org/10.1016/S0022-1694(01)00466-8.

Scherrer, S., Naef, F., Faeh, A.O., Cordery, I., 2007. Formation of runoff at the hillslope scale during intense precipitation. Hydrol. Earth Syst. Sci. 11, 907–922. https://doi.org/10.5194/hess-11- 907-2007.

Schimmelpfennig, I., Schaefer, J.M., Akçar, N., Koffman, T., Ivy-ochs, S., Schwartz, R., Finkel, R.C., Zimmerman, S., Schlüchter, C., 2014. A chronology of Holocene and Little Ice Age glacier culminations of the Steingletscher, Central Alps, Switzerland, based on high-sensitivity beryllium-10 moraine dating. Earth Planet. Sci. Lett. 393, 220–230. https://doi.org/10.1016/j.epsl.2014.02.046.

Semenova, E., 2019. Infiltrationsprozesse auf einer glazialen Chronosequenz – Präferentieller Fluss. Master Thesis. Albert-Ludwigs-University, Freiburg.

Sullivan, L.A., 1990. Soil organic matter, air encapsulation and water-stable aggregation. J. Soil Sci. 41, 529–534. https://doi.org/10.1111/j.1365-2389.1990.tb00084.x.

Swiss Federal Office of Topography, 2019. "Journey through time" – Map and "SWISSIMAGE Journey through time" – Aerial Images. [WWW Document]. URL < https://map.geo.admin.ch/mobile.html?lang = de&topic = ech&bgLayer = ch.swisstopo.swissimage&layers = ch.swisstopo.zeitreihen,ch.bfs.gebaeude_wohnungs_register,ch.bav.haltestellen-oev,ch.swisstopo.swisstlm3d-wanderwege,ch.swisstopo.swissimage-product&layers_visib > (accessed 1.3.19).

Thompson, S.E., Harman, C.J., Heine, P., Katul, G.G., 2010. Vegetation-infiltration relationships across climatic and soil type gradients. J. Geophys. Res. 115, 1–12. https://doi.org/10.1029/2009JG001134.

Tietje, O., Hennings, V., 1996. Accuracy of the saturated hydraulic conductivity prediction by pedo-transfer functions compared to the variability within FAO textural classes. Geoderma 69, 71–84. https://doi.org/10.1016/0016-7061(95)00050-X.

Tillmann, R.W., Scotter, D.R., Wallis, M.G., Clothier, B.E., 1989. Water repellency and its measurement by using intrinsic sorptivity. Aust. J. Soil Res. 27, 637–644. https://doi.org/10.1071/SR9890637.

Troch, P.A., Carrillo, G., Sivapalan, M., Wagener, T., Sawicz, K., 2013. Climate-vegetation-soil interactions and long-term hydrologic partitioning: signatures of catchment co-evolution. Hydrol. Earth Syst. Sci. 17, 2209–2217. https://doi.org/10.5194/hess-17-2209-2013.

Tromp-van Meerveld, H.J., McDonnell, J.J., 2006. On the interrelations between topography, soil depth, soil moisture, transpiration rates and species distribution at the hillslope scale. Adv. Water Resour. 29, 293–310. https://doi.org/10.1016/j.advwatres.2005.02.016.

Vilmundardóttir, O.K., Gísladóttir, G., Lal, R., 2014. Early stage development of selected soil properties along the proglacial moraines of Skaftafellsjökull glacier, SE-Iceland. Catena 121, 142–150. https://doi.org/10.1016/j.catena.2014.04.020.

Wang, J., Watts, D.B., Meng, Q., Zhang, Q., Wu, F., Torbert, H.A., 2016. Soil water infiltration impacted by maize (Zea mays L.) growth on sloping agricultural land of the Loess Plateau. J. Soil Water Conserv. 71, 301–309. https://doi.org/10.2489/jswc.71.4.301.

Weiler, M., 2017. Macropores and preferential flow — a love-hate relationship. Hydrol. Process. 31, 15–19. https://doi.org/10.1002/hyp.11074.

Weiler, M., Flühler, H., 2004. Inferring flow types from dye patterns in macroporous soils. Geoderma 120, 137–153. https://doi.org/10.1016/j.geoderma.2003.08.014.

Winkler, M., Lamprecht, A., Steinbauer, K., Hülber, K., Theurillat, J.-P., Breiner, F., Choler, P., Ertl, S., Bay, C., Fosaa, A.M., 2016. The rich sides of mountain summits – a pan-European view on aspect preferences of alpine plants. J. Biogeogr. 43, 2261–2273. https://doi.org/10.1111/jbi.12835.

Wösten, J., Pachepsky, Y.A., Rawls, W.J., 2001. Pedotransfer functions: bridging the gap between available basic soil data and missing soil hydraulic characteristics. J. Hydrol. 251, 123–150. https://doi.org/10.1016/S0022-1694(01)00464-4.

Wu, G., Liu, Y., Yang, Z., Cui, Z., Deng, L., Chang, X., Shi, Z., 2017. Root channels to indicate the increase in soil matrix water infiltration capacity of arid reclaimed mine soils. J. Hydrol. 546, 133–139. https://doi.org/10.1016/j.jhydrol.2016.12.047.

Young, M.H., McDonald, E.V., Caldwell, T.G., Benner, S.G., Meadows, D.G., 2004. Hydraulic properties of a desert soil chronosequence in the Mojave Desert, USA. Vadose Zo. J. 3 (3), 956–963. https://doi.org/10.2136/vzj2004.0956.

Zhang, J., Bruijnzeel, L.A., Marie, C., Tripoli, R., Asio, V.B., Van Meerveld, I.H.J., 2019. Soil physical characteristics of a degraded tropical grassland and a 'reforest': implications for runoff generation. Geoderma 333, 163–177. https://doi.org/10.1016/j.geoderma.2018.07.022.

Zhang, J., Van Meerveld, H.J.I., Tripoli, R., Bruijnzeel, L.A., 2018. Runoff response and sediment yield of a landslide-affected fire-climax grassland micro-catchment (Leyte, the Philippines) before and after passage of typhoon Haiyan. J. Hydrol. 565, 524–537. https://doi.org/10.1016/j.jhydrol.2018.08.016.

Zhao, Y., Wu, P., Zhao, S., Feng, H., 2013. Variation of soil infiltrability across a 79-year chronosequence of naturally restored grassland on the Loess Plateau. China. J. Hydrol. 504, 94–103. https://doi.org/10.1016/j.jhydrol.2013.09.039.

Ziegler, A.D., Giambelluca, T.W., Tran, L.T., Vana, T.T., Nullet, M.A., Fox, J., Duc, T., Pinthong, J., Maxwell, J.F., Evett, S., 2004. Hydrological consequences of landscape fragmentation in mountainous northern Vietnam: evidence of accelerated overland flow generation. J. Hydrol. 287, 124–146. https://doi.org/10.1016/j.jhydrol.2003.09.027.

Zimmermann, A., Schinn, D.S., Francke, T., Elsenbeer, H., Zimmermann, B., 2013. Uncovering patterns of near-surface saturated hydraulic conductivity in an overland flow-controlled landscape. Geoderma 195–196, 1–11. https://doi.org/10.1016/j.geoderma.2012.11.002.

Zimmermann, B., 2008. Spatial and temporal variability of soil saturated hydraulic conductivity in gradients of disturbance. J. Hydrol. 361, 78–95. https://doi.org/10.1016/j.jhydrol.2008.07.027.

Zimmermann, B., Elsenbeer, H., De Moraes, J.M., 2006. The influence of land-use changes on soil hydraulic properties: implications for runoff generation. For. Ecol. Manage. 222, 29–38. https://doi.org/10.1016/j.foreco.2005.10.070.

Zimmermann, B., Papritz, A., Elsenbeer, H., 2010. Asymmetric response to disturbance and recovery: changes of soil permeability under forest – pasture – forest transitions. Geoderma 159, 209–215. https://doi.org/10.1016/j.geoderma.2010.07.013.

Zwartendijk, B.W., Van Meerveld, H.J., Ghimire, C.P., Bruijnzeel, L.A., Ravelona, M., Jones, J.P.G., 2017. Rebuilding soil hydrological functioning after swidden agriculture in eastern Madagascar. Agric. Ecosyst. Environ. 239, 101–111. https://doi.org/10.1016/j.agee.2017.01.002.